Tongji Calculus, 7th Edition · Chapter 2

Derivatives and Differentials

Detailed Solutions

100 classic-method adaptations and high-quality synthesis problems

All mathematics compiled by XeTeX / LaTeX · 4:3 study edition

How to Use This Workbook

The material progresses from basic to challenging. IDs Q001–Q100 match across both languages and both document types. Every formula comes from explicit LaTeX source and is compiled by the XeTeX mathematics engine.

Open-text method adaptations

19 / 100

Classic-method variants

63 / 100

Original synthesis / diagnosis

18 / 100

Source-lineage policy

Problems are redesigned around classic methods found in open textbooks, public courses, and standard calculus teaching. Copyrighted books are used only for scope and method alignment; their wording and solutions are not copied. See SOURCES.md.

1. On the first attempt, use only the exercise workbook and show complete reasoning.
2. During correction, record the cause of each error instead of copying the final answer.
3. Redo errors after 48 hours and interleave topics one week later.

Basic **Single choice** **Foundation** **Open-text method adaptation**

Q001 · Recognizing the Definition of a Derivative

Problem

Suppose f is defined in a neighborhood of x_0. Which limit, if it exists, equals $f'(x_0)$?

A. $\lim_{\Delta x \to 0} \frac{f(x_0+\Delta x)-f(x_0)}{\Delta x}$

B. $\lim_{\Delta x \to 0} \frac{f(x_0+\Delta x)-f(x_0)}{x_0}$

C. $\lim_{x \to x_0} \frac{f(x)-f(x_0)}{x}$

D. $\lim_{\Delta x \to 0} \frac{f(x_0+\Delta x)+f(x_0)}{\Delta x}$

Answer

A

Knowledge points

- a derivative is the limit of a function increment divided by the corresponding input increment
- Δx must tend to 0
- the numerator must be $f(x_0 + \Delta x) - f(x_0)$

Reading and method choice

Compare every option with the increment definition. A valid difference quotient must use the base point x_0, the difference of the two function values, and the same input increment Δx.

Detailed derivation

1. When the input changes from x_0 to $x_0 + \Delta x$, its increment is Δx.
2. The corresponding function increment is $\Delta y = f(x_0 + \Delta x) - f(x_0)$, so the difference quotient is $\frac{\Delta y}{\Delta x}$.
3. If this quotient has a finite limit as $\Delta x \to 0$, that limit is $f'(x_0)$ by definition.
4. Option A matches the definition exactly; B and C use the wrong denominator, while D does not use a function increment.

Common pitfalls

- using x_0 or x instead of the corresponding input increment
- forgetting that the function increment is new value minus old value

Verification

For $f(x) = x^2$ and $x_0 = 1, A$ gives $\lim_{\Delta x \to 0} \frac{2\Delta x + \Delta x^2}{\Delta x} = 2$, the expected slope; the other forms do not reliably produce 2.

Method takeaway

Track one consistent increment: compare $x_0 + \Delta x$ with x_0 in the numerator and divide by Δx.

Extension

Writing $\Delta x = x - x_0$ gives the equivalent form $\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$.

Source lineage: Open-text method adaptation · derivative_definition · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.1. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Fill in the blank | Foundation | Open-text method adaptation

Q002 · Expressing Instantaneous Velocity as a Limit

Problem

A particle moves along a line with position $s = s(t)$. Its instantaneous velocity at $t = 3$ can be written as $v(3) =$ ____.

Answer

$\lim_{\Delta t \to 0} \frac{s(3+\Delta t)-s(3)}{\Delta t}$

Knowledge points

- average velocity is position increment divided by time increment
- instantaneous velocity is a limit of average velocities
- the derivative of position with respect to time

Reading and method choice

First write the average velocity from time 3 to $3 + \Delta t$, then shrink the time interval to 0. Position itself must not be confused with velocity.

Detailed derivation

1. On the time interval $[3, 3 + \Delta t]$, the time increment is Δt.
2. The corresponding position increment is $s(3 + \Delta t) - s(3)$.
3. Thus the average velocity is $\frac{s(3+\Delta t)-s(3)}{\Delta t}$ for $\Delta t \neq 0$.
4. Letting $\Delta t \to 0$ gives $v(3) = \lim_{\Delta t \to 0} \frac{s(3+\Delta t)-s(3)}{\Delta t}$ whenever the limit exists.

Common pitfalls

- using $\frac{s(3)}{3}$ as the instantaneous velocity
- writing $s(\Delta t) - s(3)$ in the numerator
- omitting the limiting process $\Delta t \to 0$

Verification

If $s(t) = 2t + 1$, the quotient is identically 2, so its limit is 2, as expected for constant velocity.

Method takeaway

An instantaneous rate is obtained by first forming an average rate over a finite interval and then shrinking the interval to zero.

Extension

Replacing 3 by an arbitrary t_0 gives the definition $v(t_0) = s'(t_0)$.

Source lineage: Open-text method adaptation · derivative_interpretations · reference IDs: mit-18.01sc-session-3-rate, openstax-calculus-v1-3.4. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic **Calculation** **Foundation** **Open-text method adaptation**

Q003 · Differentiate at a Nonzero Point from First Principles

Problem

Using only the definition of the derivative, find the derivative of $f(x) = x^2$ at $x = 2$.

Answer

$f'(2) = 4$

Knowledge points

- increment definition of a derivative
- expansion of a square
- canceling a nonzero increment before taking a limit

Reading and method choice

Insert $x_0 = 2$ into the difference quotient. Expand and factor out Δx, cancel it while $\Delta x \neq 0$, and only then let Δx approach 0.

Detailed derivation

1. By definition, $f'(2) = \lim_{\Delta x \to 0} \frac{(2+\Delta x)^2 - 2^2}{\Delta x}$.
2. Expand the numerator: $(2 + \Delta x)^2 - 4 = 4\Delta x + (\Delta x)^2$.
3. In the punctured neighborhood used by the limit, $\Delta x \neq 0$, so the quotient becomes $\frac{4\Delta x + (\Delta x)^2}{\Delta x} = 4 + \Delta x$.
4. Letting $\Delta x \to 0$ yields $f'(2) = \lim_{\Delta x \to 0}(4 + \Delta x) = 4$.

Common pitfalls

- substituting $\Delta x = 0$ at the beginning and obtaining $\frac{0}{0}$
- omitting the term $4\Delta x$
- confusing $f'(2)$ with $f(2)$

Verification

The graph $y = x^2$ rises near $x = 2$. Moreover, the symmetric quotient $\frac{f(2+h) - f(2-h)}{2h} = 4$ confirms the same value.

Method takeaway

The first-principles pattern is: form the quotient, simplify algebraically, cancel the increment, and then take the limit.

Extension

Replacing 2 by an arbitrary a gives $f'(a) = 2a$ by the same calculation.

Source lineage: Open-text method adaptation · derivative_definition · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.1. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | True/false with justification | Foundation | Open-text method adaptation

Q004 · Does Differentiability Guarantee Continuity?

Problem

Decide whether the statement is true and justify: if f is differentiable at x_0, then f is continuous at x_0.

Answer

True.

Knowledge points

- differentiability implies continuity
- factorization of a function increment
- a finite limit times an infinitesimal

Reading and method choice

To prove continuity, it is enough to show $f(x_0 + \Delta x) - f(x_0) \to 0$. Factor this increment into the difference quotient and Δx, then use the existence of the derivative.

Detailed derivation

1. Differentiability at x_0 means $\lim_{\Delta x\to 0} \frac{f(x_0+\Delta x)-f(x_0)}{\Delta x} = f'(x_0)$, and this limit is finite.
2. For $\Delta x \neq 0$, write $f(x_0 + \Delta x) - f(x_0) = \frac{f(x_0+\Delta x)-f(x_0)}{\Delta x} \cdot \Delta x$.
3. As $\Delta x \to 0$, the first factor tends to the finite number $f'(x_0)$, while the second tends to 0.
4. Hence the function increment tends to 0, so $\lim_{\Delta x\to 0} f(x_0 + \Delta x) = f(x_0)$; therefore f is continuous at x_0.

Common pitfalls

- reversing the implication and claiming that continuity guarantees differentiability
- failing to note that a derivative is a finite limit

Verification

The argument establishes the increment criterion $\Delta y \to 0$ for continuity without any extra assumption, so the statement is true.

Method takeaway

Differentiability is stronger than continuity: the function increment becomes a finite quantity times an infinitesimal.

Extension

The converse is false; for example, $f(x) = |x|$ is continuous but not differentiable at 0.

Source lineage: Open-text method adaptation · differentiability_and_one_sided_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic **Single choice** **Foundation** **Open-text method adaptation**

Q005 · Writing a Tangent Line from a Derivative

Problem

Suppose $f(2) = 5$ and $f'(2) = -3$. Which equation is the tangent line to $y = f(x)$ at $(2, 5)$?

A. $y - 5 = -3(x - 2)$

B. $y - 2 = -3(x - 5)$

C. $y - 5 = (\frac{1}{3})(x - 2)$

D. $y + 5 = -3(x + 2)$

Answer

A

Knowledge points

- a derivative is the tangent slope
- point-slope form
- coordinates of the point of tangency

Reading and method choice

The tangent must pass through $(2, 5)$, and its slope is $f'(2) = -3$. Insert these data into point-slope form.

Detailed derivation

1. The point on the curve corresponding to $x = 2$ is $(2, f(2)) = (2, 5)$.
2. By the geometric meaning of a derivative, the tangent slope is $k = f'(2) = -3$.
3. A line through (x_0, y_0) with slope k has point-slope form $y - y_0 = k(x - x_0)$.
4. Substitution gives $y - 5 = -3(x - 2)$, so A is correct.

Common pitfalls

- interchanging the coordinates of the point
- using the normal slope $-\frac{1}{k}$ for the tangent
- reversing signs in point-slope form

Verification

Setting $x = 2$ in A gives $y = 5$, so it passes through the required point, and its slope is exactly -3.

Method takeaway

A tangent line needs two pieces of information: $(x_0, f(x_0))$ and the slope $f'(x_0)$.

Extension

The normal slope is $\frac{1}{3}$, so the normal line is $y - 5 = (\frac{1}{3})(x - 2)$.

Source lineage: Open-text method adaptation · derivative_interpretations · reference IDs: mit-18.01sc-session-3-rate, openstax-calculus-v1-3.4. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Fill in the blank Foundation Open-text method adaptation

Q006 · One-Sided Derivatives of an Absolute-Value Function

Problem

Let $f(x) = |x|$. Then $f'_{-}(0) =$ ____, $f'_{+}(0) =$ ____, so f is ____ at 0 (write "differentiable" or "not differentiable").

Answer

$f'_{-}(0) = -1, f'_{+}(0) = 1,$ **not differentiable**

Knowledge points

- left and right derivatives
- piecewise form of absolute value
- equality of one-sided derivatives as the differentiability criterion

Reading and method choice

Determine the expression for $|h|$ separately on the two sides of 0. Both one-sided quotients have limits, but those limits are unequal.

Detailed derivation

1. Since $f(0) = 0$, the difference quotient is $\frac{f(h)-f(0)}{h} = \frac{|h|}{h}$.
2. As $h \to 0^-, h < 0$ and $|h| = -h$, so $\frac{|h|}{h} = -1$; hence $f'_{-}(0) = -1$.
3. As $h \to 0^+, h > 0$ and $|h| = h$, so $\frac{|h|}{h} = 1$; hence $f'_{+}(0) = 1$.
4. Because $-1 \neq 1$, the two-sided derivative does not exist, and f is not differentiable at 0.

Common pitfalls

- replacing $|h|$ by h on both sides
- checking only one one-sided derivative
- assuming that two finite one-sided derivatives are sufficient even when they differ

Verification

The graph $y = |x|$ has a corner at the origin; its left slope is -1 and its right slope is 1, matching the calculation.

Method takeaway

At a junction, check both sides and require the two one-sided derivatives to be exactly equal.

Extension

For $f(x) = |x - a|$, the same argument gives $f'_{-}(a) = -1$ and $f'_{+}(a) = 1$.

Source lineage: Open-text method adaptation · differentiability_and_one_sided_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Calculation Foundation Open-text method adaptation

Q007 · Tangent and Normal Lines from First Principles

Problem

Let $P(1, 1)$ lie on the curve $y = x^2$. Use the definition of the derivative to find the tangent and normal lines at P.

Answer

Tangent: $y = 2x - 1$; **normal:** $y = -(\frac{1}{2})x + \frac{3}{2}$

Knowledge points

- definition of a derivative
- tangent slope
- slope relation for perpendicular lines

Reading and method choice

First find the derivative at $x = 1$ from the difference quotient. The tangent slope equals the derivative, and because it is nonzero, the normal slope is its negative reciprocal.

Detailed derivation

1. By definition, $f'(1) = \lim_{h\to 0} \frac{(1+h)^2-1^2}{h}$.
2. Expand and cancel: $\frac{(1+h)^2-1}{h} = \frac{2h+h^2}{h} = 2 + h$, so $f'(1) = 2$.
3. The tangent passes through $(1, 1)$ with slope 2, hence $y - 1 = 2(x - 1)$, or $y = 2x - 1$.
4. The normal slope is $-\frac{1}{2}$, hence $y - 1 = -(\frac{1}{2})(x - 1)$, or $y = -(\frac{1}{2})x + \frac{3}{2}$.

Common pitfalls

- using $\frac{1}{2}$ instead of $-\frac{1}{2}$ for the normal slope
- using different points for the tangent and normal
- not using first principles as requested

Verification

Both lines pass through $(1, 1)$, and their slopes satisfy $2 \cdot (-\frac{1}{2}) = -1$, confirming perpendicularity.

Method takeaway

The geometric workflow is: obtain the slope from first principles, write the tangent in point-slope form, then take the negative reciprocal for the normal.

Extension

At (a, a^2), the tangent slope is 2a; when $a = 0$, the normal is the vertical line $x = 0$.

Source lineage: Open-text method adaptation · derivative_interpretations · reference IDs: mit-18.01sc-session-3-rate, openstax-calculus-v1-3.4. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Open-text method adaptation

Q008 · Choose the Derivative of a Logarithm

Problem

For $x > 0$, what is the derivative of $y = \ln x$?

A. $y' = \frac{1}{x}$

B. $y' = \ln x$

C. $y' = x$

D. $y' = e^x$

Answer

A

Knowledge points

- basic derivative formula for the natural logarithm
- domain of applicability
- distinguishing a function from its derivative

Reading and method choice

Apply the basic formula $(\ln x)' = \frac{1}{x}$ and retain the stated domain $x > 0$.

Detailed derivation

1. The real-valued natural logarithm $\ln x$ has domain $x > 0$.
2. The basic derivative formula is $\frac{d(\ln x)}{dx} = \frac{1}{x}$.
3. Therefore $y' = \frac{1}{x}$, which is defined throughout $x > 0$.
4. Thus A is correct; B repeats the original function, while C and D belong to different formulas.

Common pitfalls

- writing $(\ln x)' = \ln x$
- ignoring the domain $x > 0$
- confusing the formula with $(e^x)' = e^x$

Verification

At $x = 1$ the formula gives $y' = 1$, which agrees with the difference-quotient limit of $\ln x$ at 1.

Method takeaway

Learn each basic formula together with the function, its derivative, and its domain.

Extension

The chain rule gives $(\ln u(x))' = \frac{u'(x)}{u(x)}$, provided $u(x) > 0$.

Source lineage: Open-text method adaptation · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.9. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Classic-method variant

Q009 · A Polynomial Times an Exponential

Problem

Let $f(x) = x^2e^x$. Which expression equals $f'(x)$?

A. $2xe^x$

B. $e^x(x^2 + 2x)$

C. x^2e^x

D. $e^x(x + 2)$

Answer

B. $e^x(x^2 + 2x)$**.**

Knowledge points

- Product rule
- $(e^x)' = e^x$
- Factoring a common factor

Reading and method choice

The function is the product of x^2 and e^x. Differentiate each factor in turn while retaining the other factor, then combine the terms.

Detailed derivation

1. Set $u(x) = x^2$ and $v(x) = e^x$, so $f(x) = u(x)v(x)$.
2. Differentiate the factors: $u'(x) = 2x$ and $v'(x) = e^x$.
3. Apply $f' = u'v + uv'$ to obtain $f'(x) = 2\text{xe}^x + x^2e^x$.
4. Factor out e^x to get $f'(x) = e^x(x^2 + 2x)$.
5. Therefore the correct choice is B.

Common pitfalls

- Writing $u'v'$ instead of the product rule
- Differentiating only one factor
- Losing a factor of x when factoring

Verification

At $x = 0$, the definition gives $\lim_{h\to 0} \frac{h^2 e^h}{h} = 0$, and the derived formula also gives 0.

Method takeaway

The derivative of a product is a sum of two terms, not the product of two derivatives.

Extension

Replace x^2 by x^m and verify the pattern $(x^m e^x)' = e^x(x^m + mx^{m-1})$.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.9. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | True/false with justification | Foundation | Original synthesis / diagnosis

Q010 · Can a Quotient Be Differentiated Termwise?

Problem

True or false, with justification: If u and v are differentiable and $v'(x) \neq 0$, then $[\frac{u(x)}{v(x)}]' = \frac{u'(x)}{v'(x)}$.

Answer

False. The needed condition is $v(x) \neq 0$, and the correct formula is $(\frac{u}{v})' = \frac{u'v-uv'}{v^2}$.

Knowledge points

- Quotient rule
- Conditions for a formula
- Counterexample testing

Reading and method choice

The statement has both an incorrect formula and the wrong nonzero condition. A simple counterexample disproves it, after which the correct rule can be stated.

Detailed derivation

1. For $\frac{u}{v}$ to be defined and the quotient rule to apply locally, the condition is $v(x) \neq 0$, not $v'(x) \neq 0$.
2. The correct formula is $(\frac{u}{v})' = \frac{u'v-uv'}{v^2}$.
3. Choose $u(x) = x$ and $v(x) = x + 1$ on an interval excluding $x = -1$.
4. The proposed formula gives $\frac{u'}{v'} = 1$, whereas $(\frac{x}{x+1})' = \frac{(x+1)-x}{(x+1)^2} = \frac{1}{(x+1)^2}$.
5. These are not generally equal, so the statement is false.

Common pitfalls

- Treating a quotient as the quotient of derivatives
- Replacing $v \neq 0$ by $v' \neq 0$
- Forgetting the minus sign in the numerator

Verification

At $x = 1$, the actual derivative is $\frac{1}{4}$ while the incorrect formula gives 1.

Method takeaway

The quotient rule is denominator times numerator derivative minus numerator times denominator derivative, divided by the denominator squared.

Extension

If v is a nonzero constant, the quotient rule does reduce to $(\frac{u}{v})' = \frac{u'}{v}$.

Source lineage: Original synthesis / diagnosis · elementary_differentiation_rules · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.3. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Fill in the blank | Foundation | Classic-method variant

Q011 · A Linear Combination of Three Basic Functions

Problem

For $x > 0$, if $y = 3e^x - 2\ln x + 5\arctan x$, then $y' =$ ____.

Answer

$y' = 3e^x - \frac{2}{x} + \frac{5}{1+x^2}$.

Knowledge points

- Linearity of differentiation
- $(\ln x)' = \frac{1}{x}$
- $(\arctan x)' = \frac{1}{1+x^2}$

Reading and method choice

This is a linear combination of exponential, logarithmic, and inverse-tangent functions, so the terms can be differentiated separately. The condition $x > 0$ also ensures that $\ln x$ is defined.

Detailed derivation

1. By the constant-multiple rule, $(3e^x)' = 3e^x$.
2. For $x > 0, (-2\ln x)' = -\frac{2}{x}$.
3. By the inverse-tangent formula, $(5\arctan x)' = \frac{5}{1+x^2}$.
4. Add the three derivatives to obtain $y' = 3e^x - \frac{2}{x} + \frac{5}{1+x^2}$.
5. The resulting expression is defined everywhere on the original domain $x > 0$.

Common pitfalls

- Writing $(\ln x)'$ as $\ln x$
- Omitting $1 + x^2$ in the inverse-tangent derivative
- Ignoring the domain of $\ln x$

Verification

Setting any two coefficients to zero reduces the result to the corresponding basic derivative formula.

Method takeaway

Sums, differences, and constant multiples differentiate termwise, but each basic formula and its domain must be respected.

Extension

If $\ln x$ is replaced by $\ln |x|$, the same derivative formula holds for $x \neq 0$.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.5, openstax-calculus-v1-3.7, openstax-calculus-v1-3.9. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Calculation | Foundation | Open-text method adaptation

Q012 · The Fifth Power of a Linear Function

Problem

Differentiate $y = (2x - 1)^5$ and explicitly identify the outer and inner functions.

Answer

$y' = 10(2x - 1)^4$.

Knowledge points

- Chain rule
- Layering a composite function
- Power rule

Reading and method choice

Treat $2x - 1$ as one unit. The outer function is $u \mapsto u^5$ and the inner function is $x \mapsto 2x - 1$, so the derivative is the outer derivative times the inner derivative.

Detailed derivation

1. Introduce the inner variable $u = 2x - 1$, so $y = u^5$.
2. Differentiate the outer function: $\frac{dy}{du} = 5u^4$.
3. Differentiate the inner function: $\frac{du}{dx} = 2$.
4. The chain rule gives $y' = (\frac{dy}{du})(\frac{du}{dx}) = 5u^4 \cdot 2$.
5. Substitute $u = 2x - 1$ to obtain $y' = 10(2x - 1)^4$.

Common pitfalls

- Writing only $5(2x - 1)^4$ and missing the factor 2
- Applying the exponent to the inner derivative
- Expanding unnecessarily and creating algebra errors

Verification

The leading term of the expansion is $32x^5$, whose derivative has leading term $160x^4$; the result has leading term $10 \cdot 16x^4 = 160x^4$.

Method takeaway

Each layer passed through in a composite function contributes its own derivative factor.

Extension

For $y = (ax + b)^n$, the same reasoning gives $y' = an(ax + b)^{n-1}$.

Source lineage: Open-text method adaptation · chain_rule_and_composites · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.6. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Calculation Foundation Classic-method variant

Q013 · A Sine Divided by an Exponential

Problem

Differentiate $y = \frac{\sin x}{e^x}$ and write the result in the simplest product form.

Answer

$y' = e^{-x}(\cos x - \sin x)$.

Knowledge points

- Quotient rule
- Rewriting with a negative exponent
- Derivatives of sine and exponential functions

Reading and method choice

One may apply the quotient rule directly or rewrite the function as $e^{-x} \sin x$ and use the product rule. The second route produces a compact form naturally.

Detailed derivation

1. Rewrite the function as $y = e^{-x} \sin x$.
2. By the chain rule, $(e^{-x})' = -e^{-x}$, while $(\sin x)' = \cos x$.
3. Apply the product rule: $y' = (-e^{-x}) \sin x + e^{-x} \cos x$.
4. Factor e^{-x} to obtain $y' = e^{-x}(\cos x - \sin x)$.
5. Because e^x is never zero, the formula holds for every real x.

Common pitfalls

- Writing $(e^{-x})'$ as e^{-x}
- Reversing the subtraction in the quotient rule
- Losing the minus sign while simplifying

Verification

The quotient rule gives $\frac{e^x \cos x - e^x \sin x}{e^{2x}} = \frac{\cos x - \sin x}{e^x}$, which is identical.

Method takeaway

When an exponential appears in the denominator, rewriting it with a negative exponent often simplifies differentiation.

Extension

Replacing $\sin x$ by a differentiable $f(x)$ gives $[e^{-x}f(x)]' = e^{-x}[f'(x) - f(x)]$.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.5, openstax-calculus-v1-3.9. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Classic-method variant

Q014 · The Derivative of an Inverse at Corresponding Points

Problem

Suppose f has a differentiable inverse $g = f^{-1}$ near 2, with $f(2) = 5$ and $f'(2) = -3$. What is $g'(5)$?

A. -3

B. $-\frac{1}{3}$

C. $\frac{1}{3}$

D. 3

Answer

B. $-\frac{1}{3}$.

Knowledge points

- Inverse-function derivative formula
- Corresponding points of a function and its inverse
- Nonzero derivative condition

Reading and method choice

First use $f(2) = 5$ to identify $g(5) = 2$, then take the reciprocal of the original derivative at this corresponding point.

Detailed derivation

1. Since $g = f^{-1}$ and $f(2) = 5$, we have $g(5) = 2$.
2. The inverse derivative formula is $g'(y) = \frac{1}{f'(g(y))}$.
3. At $y = 5, g'(5) = \frac{1}{f'(g(5))}$.
4. Substitute $g(5) = 2$ and $f'(2) = -3$ to obtain $g'(5) = \frac{1}{-3} = -\frac{1}{3}$.
5. Therefore the correct choice is B.

Common pitfalls

- Using $f'(2)$ itself as $g'(5)$
- Using 2 as the inverse function's input

- Dropping the reciprocal or the negative sign

Verification

The product of the corresponding slopes is $f'(2)g'(5) = (-3)(-\frac{1}{3}) = 1$, consistent with reflection of inverse graphs across $y = x$.

Method takeaway

At corresponding points, the derivative of an inverse is the reciprocal of the original derivative.

Extension

If $f'(2) = 0$, the formula cannot be used to claim that g has a finite derivative at 5.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.7. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | True/false with justification | Foundation | Classic-method variant

Q015 · Higher Derivatives of a Polynomial

Problem

True or false, with justification: If $y = x^4 - 3x + 2$, then $y^{(5)}(x) \equiv 0$.

Answer

True. Every derivative of order five or higher of a fourth-degree polynomial is identically 0.

Knowledge points

- Definition of higher derivatives
- Degree reduction for polynomials
- Identically zero functions

Reading and method choice

Each differentiation lowers the polynomial degree by one. The linear term disappears after two derivatives, and the fourth-degree term disappears after five.

Detailed derivation

1. The first derivative is $y' = 4x^3 - 3$.
2. The second derivative is $y'' = 12x^2$.
3. The third and fourth derivatives are $y''' = 24x$ and $y^{(4)} = 24$.
4. Differentiating once more gives $y^{(5)} = 0$ because the derivative of 24 is zero.
5. Further derivatives of the zero function remain zero, so the assertion holds identically.

Common pitfalls

- Treating the fifth derivative as $5x^{-1}$
- Forgetting that lower-degree terms disappear earlier
- Confusing a zero at one point with an identically zero function

Verification

The formula $(x^4)^{(n)} = \frac{4!}{(4-n)!} \cdot x^{4-n}$ applies only for $n \leq 4$; at $n = 5$ the derivative is already zero.

Method takeaway

Every derivative of order $m + 1$ or higher of a degree-m polynomial is identically zero.

Extension

If the leading coefficient of a degree-m polynomial is nonzero, its mth derivative is the nonzero constant $m!a_m$.

Source lineage: Classic-method variant · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic **Fill in the blank** **Foundation** **Classic-method variant**

Q016 · The 2026th Derivative of Cosine

Problem

If $y = \cos x$, then $y^{(2026)} =$ ____.

Answer

$y^{(2026)} = -\cos x.$

Knowledge points

- Higher derivatives of cosine
- Classification modulo 4
- Cyclic patterns

Reading and method choice

Successive derivatives of cosine repeat every four steps, so only the remainder when 2026 is divided by 4 matters.

Detailed derivation

1. The first four derivatives are $y' = -\sin x$, $y'' = -\cos x$, $y''' = \sin x$, and $y^{(4)} = \cos x$.
2. The fourth derivative returns to the original function, so the sequence has period 4.
3. Since $2026 = 4 \cdot 506 + 2$, the 2026th derivative occupies the same position as the second derivative.
4. Because $y'' = -\cos x$, we obtain $y^{(2026)} = -\cos x$.
5. This identity holds for every real x.

Common pitfalls

- Computing the remainder of 2026 as 0
- Dropping the minus sign in the second derivative
- Assuming the period is 2

Verification

Because 2024 is divisible by 4, $y^{(2024)} = \cos x$; differentiating twice more indeed gives $-\cos x$.

Method takeaway

For very high trigonometric derivatives, find the cycle first and then use modular arithmetic.

Extension

A unified form is $(\cos x)^{(n)} = \cos(x + \frac{n\pi}{2})$.

Source lineage: Classic-method variant · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic **Single choice** **Foundation** **Open-text method adaptation**

Q017 · Recognizing a First Implicit Derivative

Problem

On the curve $x^2 + xy + y^2 = 7$, wherever y is a differentiable function of x, which expression equals y'?

A. $-\frac{2x+y}{x+2y}$

B. $-\frac{2x+y}{2x+y}$

C. $\frac{2x+y}{x+2y}$

D. $-\frac{x+2y}{2x+y}$

Answer

A

Knowledge points

- Differentiating both sides with respect to x
- Product rule for xy
- Collecting terms containing y'

Reading and method choice

Treat y as $y(x)$. Differentiating xy and y^2 both produces y' terms, which must then be collected.

Detailed derivation

1. Differentiate x^2 to obtain 2x.
2. The product rule gives $\frac{d(xy)}{dx} = y + xy'$, and the chain rule gives $\frac{d(y^2)}{dx} = 2yy'$.
3. The derivative of 7 is 0, so $2x + y + xy' + 2yy' = 0$.
4. Collecting y' gives $(x + 2y)y' = -(2x + y)$.
5. When $x + 2y \neq 0, y' = -\frac{2x+y}{x+2y}$, so A is correct.

Common pitfalls

- Writing $\frac{d(xy)}{dx}$ as only xy'
- Omitting y' from $\frac{d(y^2)}{dx}$
- Losing the minus sign while rearranging

Verification

Substituting A into $(x + 2y)y' = -(2x + y)$ satisfies the equation identically; the other options generally do not.

Method takeaway

In implicit differentiation, regard every y as $y(x)$ and collect all y' terms at the end.

Extension

If $x + 2y = 0$, return to the local geometry to check for a vertical tangent rather than using the fraction.

Source lineage: Open-text method adaptation · implicit_differentiation · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.8. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Calculation Foundation Open-text method adaptation

Q018 · The Tangent to a Circle

Problem

At the point $(3, 4)$ on $x^2 + y^2 = 25$, find $\frac{dy}{dx}$ and the equation of the tangent line.

Answer

$\frac{dy}{dx} = -\frac{3}{4}$; the tangent is $y - 4 = -(\frac{3}{4})(x - 3)$, equivalently $3x + 4y = 25$.

Knowledge points

- Implicit differentiation of a circle
- Point-slope equation
- Substitution into the slope

Reading and method choice

First differentiate the circle implicitly to obtain the general slope, then substitute the point and use point-slope form.

Detailed derivation

1. Differentiate $x^2 + y^2 = 25$ with respect to x to get $2x + 2yy' = 0$.
2. When $y \neq 0$, solve for $y' = -\frac{x}{y}$.
3. At $(3, 4)$, $y' = -\frac{3}{4}$.
4. Point-slope form gives $y - 4 = -(\frac{3}{4})(x - 3)$.
5. Multiplying by 4 and rearranging gives the equivalent form $3x + 4y = 25$.

Common pitfalls

- Using the positive slope $\frac{3}{4}$
- Omitting y' when differentiating y^2
- Writing a tangent that does not pass through the point

Verification

The line $3x + 4y = 25$ contains $(3, 4)$ because $9 + 16 = 25$, and its slope is $-\frac{3}{4}$.

Method takeaway

The implicit derivative is the local tangent slope; then apply point-slope form.

Extension

The normal slope is $\frac{4}{3}$, so the normal is $y - 4 = (\frac{4}{3})(x - 3)$.

Source lineage: Open-text method adaptation · implicit_differentiation · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.8. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Fill in the blank | Foundation | Classic-method variant

Q019 · Evaluating an Implicit Derivative at a Point

Problem

For $x^2 + xy = 6$, $\frac{dy}{dx}$ at $(2, 1)$ is ____.

Answer

$-\frac{5}{2}$

Knowledge points

- Product rule
- First implicit derivative
- Point substitution

Reading and method choice

Because xy contains both x and $y(x)$, the product rule is required. Solve the derivative relation before substituting the point.

Detailed derivation

1. Differentiate x^2 to get 2x.
2. Differentiate xy to get $y + xy'$.
3. The derivative of 6 is 0, so $2x + y + xy' = 0$.
4. At $(2, 1)$, this becomes $4 + 1 + 2y' = 0$.
5. Thus $2y' = -5$ and $y' = -\frac{5}{2}$.

Common pitfalls

- Writing the derivative of xy as y'
- Substituting the point before differentiating
- Forgetting to divide by 2

Verification

The general formula is $y' = -\frac{2x+y}{x}$; at $x = 2, y = 1$ it gives $-\frac{5}{2}$.

Method takeaway

Derive the general implicit relation first, then substitute the point.

Extension

Near a point with $x = 0$, the displayed formula has zero denominator and one must recheck whether y is locally a function of x.

Source lineage: Classic-method variant · implicit_differentiation · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.8. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | True/false with justification | Foundation | Classic-method variant

Q020 · The Domain of an Implicit-Derivative Formula

Problem

True or false, with justification: From $x^2 + y^2 = 1$ we obtain $y' = -\frac{x}{y}$, so this formula gives a finite tangent slope at every point of the circle.

Answer

False. At $(\pm 1, 0), y = 0$, the curve has a vertical tangent, and y is not locally a single-valued differentiable function of x with a finite derivative.

Knowledge points

- Denominator condition in implicit differentiation
- Vertical tangents
- Local functional representation

Reading and method choice

The relation $2x + 2yy' = 0$ is valid, but solving it as $y' = -\frac{x}{y}$ introduces the condition $y \neq 0$.

Detailed derivation

1. Differentiate the circle to obtain $2x + 2yy' = 0$.
2. Division by 2y is allowed only when $y \neq 0$, yielding $y' = -\frac{x}{y}$.
3. The points $(1, 0)$ and $(-1, 0)$ lie on the circle and have $y = 0$, so the fraction is undefined there.
4. The tangent near $(1, 0)$ is $x = 1$, and near $(-1, 0)$ it is $x = -1$; both are vertical lines.
5. A vertical line has no finite slope, so the statement is false.

Common pitfalls

- Ignoring a possible zero divisor
- Confusing a nonexistent finite derivative with no tangent
- Assuming every implicit curve globally gives y as a single-valued function of x

Verification

At $(1,0)$ the radius is horizontal, so the tangent is vertical, agreeing with the zero denominator.

Method takeaway

An implicit-derivative formula must be used with its nonzero-denominator condition; a zero denominator often signals a vertical tangent.

Extension

If x is viewed as a function of y, then at $(\pm 1,0)$ one has $\frac{dx}{dy} = -\frac{y}{x} = 0$.

Source lineage: Classic-method variant · implicit_differentiation · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.8. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Open-text method adaptation

Q021 · Recognizing the Differential of a Function

Problem

If $y = f(x)$ is differentiable at x and $dx = \Delta x$, which expression is the differential dy?

A. $dy = f'(x)dx$

B. $dy = \Delta y$ for every finite Δx

C. $dy = f(x)dx$

D. dy is independent of dx

Answer

A

Knowledge points

- Definition of a one-variable differential
- Independent-variable differential dx
- Difference between increment and differential

Reading and method choice

For a differentiable function, the increment is a linear principal part plus a higher-order small term. The linear part $f'(x)\Delta x$ is defined as dy.

Detailed derivation

1. Let the independent-variable increment be Δx and set $dx = \Delta x$.
2. Differentiability gives $\Delta y = f'(x)\Delta x + o(\Delta x)$.
3. The part $f'(x)\Delta x$ that is linear in Δx is defined as the differential.
4. Therefore $dy = f'(x)dx$, so A is correct.
5. In general dy only approximates Δy; it is not exactly equal for every finite increment.

Common pitfalls

- Treating dy and Δy as always identical
- Using $f(x)$ instead of $f'(x)$

- Assuming dy is independent of dx

Verification

For $f(x) = x^2, \Delta y = 2x\Delta x + (\Delta x)^2$ while $dy = 2xdx$; their difference is $(\Delta x)^2$.

Method takeaway

The differential is the part of the function increment that is linear in the independent-variable increment.

Extension

For a linear function $y = ax + b$, the higher-order remainder is zero, so $dy = \Delta y$ exactly for every increment.

Source lineage: Open-text method adaptation · differentials_and_linear_approximation · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Fill in the blank Foundation Classic-method variant

Q022 · Numerical Differential of a Power Function

Problem

If $y = x^3$, then at $x = 2$ with $dx = 0.01, dy =$ ____.

Answer

0.12

Knowledge points

- $dy = f'(x)dx$
- Power-function derivative
- Numerical substitution

Reading and method choice

First obtain $dy = 3x^2dx$, then substitute the base point $x = 2$ and $dx = 0.01$.

Detailed derivation

1. Differentiate $y = x^3$ to get $y' = 3x^2$.
2. The differential is $dy = y'dx = 3x^2dx$.
3. At $x = 2, 3x^2 = 3 \cdot 4 = 12$.
4. With $dx = 0.01$, substitute into $dy = 12dx$ to obtain $dy = 12 \cdot 0.01$.
5. Therefore $dy = 0.12$; this is the linear estimate of the corresponding output increment.

Common pitfalls

- Evaluating the derivative at $x + dx = 2.01$
- Writing $d(x^3) = x^2dx$
- Forgetting to multiply by dx

Verification

The exact increment is $2.01^3 - 2^3 = 0.120601$. The differential 0.12 is close but not identical.

Method takeaway

Evaluate the differential coefficient at the base point, then multiply by the small input change.

Extension

If $dx = -0.01$, then $dy = -0.12$, reflecting the local direction of change.

Source lineage: Classic-method variant · differentials_and_linear_approximation · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Open-text method adaptation

Q023 · Derivative of the Reciprocal Function at Any Nonzero Point

Problem

Let $f(x) = \frac{1}{x}$. For any $a \neq 0$, use only the definition of the derivative to find $f'(a)$.

Answer

$f'(a) = -\frac{1}{a^2}$

Knowledge points

- derivative definition at a nonzero point
- simplifying a rational difference quotient
- domain control in a limit

Reading and method choice

Write the difference quotient at a and combine the fractions. Since $a \neq 0$ and $a + h \neq 0$ for all sufficiently small h, the algebraic simplification is legitimate.

Detailed derivation

1. By definition, $f'(a) = \lim_{h \to 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h}$.
2. Combine the numerator: $\frac{1}{a+h} - \frac{1}{a} = \frac{a-(a+h)}{a(a+h)} = -\frac{h}{a(a+h)}$.
3. For $h \neq 0$, the quotient simplifies to $\frac{-\frac{h}{a(a+h)}}{h} = -\frac{1}{a(a+h)}$.
4. Let $h \to 0$. Since $a \neq 0, f'(a) = -\frac{1}{a \cdot a} = -\frac{1}{a^2}$.

Common pitfalls

- reversing the sign when combining the fractions
- losing the minus sign after canceling h
- omitting the conditions $a \neq 0$ and $a + h \neq 0$

Verification

At $a = 1$ the result is -1. As x increases slightly from 1, $\frac{1}{x}$ decreases, which agrees with the negative derivative.

Method takeaway

For rational functions, combining fractions often exposes the input increment that can be canceled.

Extension

The same method proves $(\frac{c}{x})' = -\frac{c}{x^2}$ for any constant c.

Source lineage: Open-text method adaptation · derivative_definition · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.1. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Multiple choice | Methods | Original synthesis / diagnosis

Q024 · Logical Relations Involving Differentiability

Problem

Let x_0 be an interior point of the domain of f. Which statements are correct?

A. If $f'(x_0)$ exists, then $f'_-(x_0)$ and $f'_+(x_0)$ both exist and are equal

B. If f is continuous at x_0, then $f'(x_0)$ must exist

C. If $f'(x_0)$ exists, then f is continuous at x_0

D. If $f'_-(x_0)$ and $f'_+(x_0)$ are both finite, then $f'(x_0)$ must exist

Answer

A, C

Knowledge points

- relation between two-sided and one-sided derivatives
- differentiability implies continuity
- testing claims by counterexample

Reading and method choice

For A and D, equality of the one-sided derivatives is decisive. For B and C, identify the correct direction of the implication from differentiability to continuity. The function $|x - x_0|$ rejects both common false claims.

Detailed derivation

1. If the two-sided difference-quotient limit $f'(x_0)$ exists, its left and right limits exist and equal that same value; thus A is true.
2. Differentiability implies continuity, so C is true.
3. The function $f(x) = |x - x_0|$ is continuous at x_0, but its left and right derivatives are -1 and 1; hence B is false.
4. In the same example both one-sided derivatives are finite but unequal, so the two-sided derivative does not exist; hence D is false.

Common pitfalls

- interchanging a necessary condition and a sufficient condition
- requiring the one-sided derivatives to be finite but not equal
- treating continuity and differentiability as equivalent

Verification

For $|x - x_0|$, continuity at x_0 is immediate, while the left and right quotients tend exactly to -1 and 1, validating the counterexample.

Method takeaway

At an interior point, differentiability is equivalent to existence and equality of both one-sided derivatives; continuity is only necessary.

Extension

At an endpoint of the domain, one normally studies the one-sided derivative from within the domain rather than a two-sided derivative.

Source lineage: Original synthesis / diagnosis · differentiability_and_one_sided_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Proof Methods Open-text method adaptation

Q025 · Prove Differentiability Implies Continuity via an Infinitesimal Remainder

Problem

Suppose f is differentiable at x_0. Prove that some $\alpha(\Delta x) \to 0$ satisfies $f(x_0 + \Delta x) - f(x_0) = f'(x_0)\Delta x + \alpha(\Delta x)\Delta x$, and use it to prove continuity at x_0.

Answer

For $\Delta x \neq 0$, set $\alpha(\Delta x) = \frac{f(x_0+\Delta x)-f(x_0)}{\Delta x} - f'(x_0)$, and set $\alpha(0) = 0$. This gives the representation and continuity.

Knowledge points

- increment form of differentiability
- infinitesimal remainder
- increment criterion for continuity

Reading and method choice

Start from the difference quotient tending to $f'(x_0)$, and name its error $\alpha(\Delta x)$. This error tends to 0; multiplying back by Δx gives the linear principal part of the function increment.

Detailed derivation

1. Because f is differentiable at x_0, $\lim_{\Delta x \to 0} \frac{f(x_0+\Delta x)-f(x_0)}{\Delta x} = f'(x_0)$.
2. For $\Delta x \neq 0$ define $\alpha(\Delta x) = \frac{f(x_0+\Delta x)-f(x_0)}{\Delta x} - f'(x_0)$, and complete the definition by setting $\alpha(0) = 0$.
3. The derivative limit immediately gives $\alpha(\Delta x) \to 0$. Rearranging and multiplying by Δx yields $f(x_0+\Delta x) - f(x_0) = f'(x_0)\Delta x + \alpha(\Delta x)\Delta x$.
4. As $\Delta x \to 0$, both $f'(x_0)\Delta x$ and $\alpha(\Delta x)\Delta x$ tend to 0, so the function increment tends to 0 and f is continuous at x_0.

Common pitfalls

- stating the increment formula without constructing α
- failing to prove $\alpha(\Delta x) \to 0$
- treating $\alpha(\Delta x)\Delta x$ as a constant independent of Δx

Verification

Dividing the representation by Δx for $\Delta x \neq 0$ recovers the quotient $f'(x_0) + \alpha(\Delta x)$, whose limit is indeed $f'(x_0)$.

Method takeaway

Differentiability provides more than continuity: the function increment has a linear principal part $f'(x_0)\Delta x$ and a smaller remainder.

Extension

This representation directly motivates the later definition of a differential through $\Delta y = A\Delta x + o(\Delta x)$.

Source lineage: Open-text method adaptation · differentiability_and_one_sided_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Calculation** **Methods** **Classic-method variant**

Q026 · Matching Parameters at a Junction

Problem

Let $f(x) = \begin{cases} x^2, & x \le 1, \\ ax+b, & x > 1. \end{cases}$ Find a, b so that f is differentiable at $x = 1$, and find $f'(1)$.

Answer

$a = 2, b = -1, f'(1) = 2$

Knowledge points

- continuity at a piecewise junction
- equality of one-sided derivatives
- parameter equations

Reading and method choice

Differentiability first requires continuity and then equality of the one-sided derivatives. Continuity determines $a + b$, while slope matching determines a.

Detailed derivation

1. From the left branch and the definition, $f(1) = 1$; the right-hand limit is $\lim_{x \to 1^+}(ax + b) = a + b$.
2. Continuity requires $a + b = 1$, which is necessary for differentiability.
3. The left derivative is $\lim_{h \to 0^-} \frac{(1+h)^2 - 1}{h} = 2$. Using $a + b = 1$, the right derivative is $\lim_{h \to 0^+} \frac{a(1+h)+b-1}{h} = a$.
4. Equality of the one-sided derivatives gives $a = 2$; then $a + b = 1$ gives $b = -1$, and $f'(1) = 2$.

Common pitfalls

- matching function values but not slopes
- finding the right derivative without first using $a + b = 1$, leaving a constant term in the quotient
- writing $a - b = 1$ instead of $a + b = 1$

Verification

With $a = 2$ and $b = -1$, the right branch is $2x - 1$, which has value 1 and slope 2 at $x = 1$, matching the left branch x^2.

Method takeaway

Differentiability at a piecewise junction requires two matches: function values first, then one-sided slopes.

Extension

If only continuity were required, every pair satisfying $a + b = 1$ would work; $a = 2$ would not be necessary.

Source lineage: Classic-method variant · differentiability_and_one_sided_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Fill in the blank | Methods | Classic-method variant

Q027 · Instantaneous Velocity at a Specified Time

Problem

A particle has position $s(t) = t^3 - 6t^2 + 9t$ meters, with t in seconds. Use a difference-quotient limit to find the instantaneous velocity at $t = 2$, namely $v(2) =$ ____.

Answer

$-3\,\mathrm{m\,s^{-1}}$

Knowledge points

- derivative of a position function
- instantaneous velocity
- directional meaning of a negative sign

Reading and method choice

Compute $\frac{s(2+h)-s(2)}{h}$ from the definition. After expansion, all constant terms should cancel, leaving a factor h.

Detailed derivation

1. First, $s(2) = 8 - 24 + 18 = 2$.
2. Expand $s(2+h) = (2+h)^3 - 6(2+h)^2 + 9(2+h) = 2 - 3h + h^3$.
3. The average-velocity quotient is $\frac{s(2+h)-s(2)}{h} = \frac{-3h+h^3}{h} = -3 + h^2$.
4. Letting $h \to 0$ gives $v(2) = \lim_{h\to 0}(-3 + h^2) = -3\,\mathrm{m\,s^{-1}}$; the negative sign indicates motion opposite the chosen positive direction.

Common pitfalls

- using the position $s(2) = 2\,\mathrm{m}$ as the velocity
- omitting cross terms in the expansion
- dropping the velocity unit or misreading the negative sign

Verification

The basic rules also give $s'(t) = 3t^2 - 12t + 9$, and substitution $t = 2$ gives -3, matching the first-principles result.

Method takeaway

Instantaneous velocity is the derivative of position with respect to time; its sign records direction, not merely speed.

Extension

The average velocity near $t = 2$ is $-3 + h^2$, which approaches $-3\,\mathrm{m\,s^{-1}}$ for small h.

Source lineage: Classic-method variant · derivative_interpretations · reference IDs: mit-18.01sc-session-3-rate, openstax-calculus-v1-3.4. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q028 · Tangent and Normal to a Square-Root Curve

Problem

Without using a ready-made derivative formula, find the derivative of $y = \sqrt{x}$ at $(4, 2)$ from the definition, and write the tangent and normal lines.

Answer

$f'(4) = \frac{1}{4}$; **tangent:** $y - 2 = (\frac{1}{4})(x - 4)$; **normal:** $y - 2 = -4(x - 4)$

Knowledge points

- rationalizing a radical difference quotient
- geometric meaning of a derivative
- normal slope as the negative reciprocal

Reading and method choice

The quotient contains two radicals, so multiply by the conjugate to cancel h. The derivative is nonzero, so the normal slope exists and equals $-\frac{1}{f'(4)}$.

Detailed derivation

1. By definition, $f'(4) = \lim_{h\to 0} \frac{\sqrt{4+h}-2}{h}$.
2. Rationalizing gives $\frac{(4+h)-4}{h[\sqrt{4+h}+2]} = \frac{1}{\sqrt{4+h}+2}$.
3. Let $h \to 0$ to obtain $f'(4) = \frac{1}{2+2} = \frac{1}{4}$; hence the tangent is $y - 2 = (\frac{1}{4})(x - 4)$.
4. The normal slope is -4, so the normal line is $y - 2 = -4(x - 4)$.

Common pitfalls

- multiplying only the numerator by the conjugate
- treating the limit of $\sqrt{4 + h}$ as 4
- omitting the minus sign in the normal slope

Verification

Both lines pass through $(4, 2)$, their slopes multiply to $(\frac{1}{4})(-4) = -1$, and the positive tangent slope agrees with the curve rising there.

Method takeaway

Use the conjugate for a radical difference quotient, then convert the derivative into geometric line equations.

Extension

The same rationalization gives $(\sqrt{x})' = \frac{1}{2\sqrt{x}}$ for $x > 0$.

Source lineage: Classic-method variant · derivative_interpretations · reference IDs: mit-18.01sc-session-3-rate, openstax-calculus-v1-3.4. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | True/false with justification | Methods | Classic-method variant

Q029 · Testing the Converse of Differentiability Implies Continuity

Problem

Decide whether the statement is true and justify rigorously: if f is continuous at x_0, then f is differentiable at x_0.

Answer

False; for example, $f(x) = |x - x_0|$ is continuous but not differentiable at x_0.

Knowledge points

- necessary versus sufficient conditions
- constructing a counterexample
- one-sided derivative test

Reading and method choice

One counterexample disproves a universal statement. Choose an absolute-value function with a corner at x_0, and verify both continuity and failure of differentiability.

Detailed derivation

1. Let $f(x) = |x - x_0|$, so $f(x_0) = 0$.
2. As $x \to x_0$, $|x - x_0| \to 0 = f(x_0)$, so f is continuous at x_0.
3. For $h \to 0^-$, the quotient $\frac{|h|}{h} = -1$, so $f'_-(x_0) = -1$.
4. For $h \to 0^+$, the quotient $\frac{|h|}{h} = 1$, so $f'_+(x_0) = 1$. The values differ, so f is not differentiable at x_0 and the statement is false.

Common pitfalls

- drawing a graph without calculating the one-sided derivatives
- turning the one-way implication into an equivalence
- giving a counterexample without checking continuity

Verification

The construction works for every prescribed x_0, so it genuinely disproves the universal claim.

Method takeaway

Continuity excludes jumps but not corners; it is necessary, not sufficient, for differentiability.

Extension

The function $f(x) = |x - x_0|^{\frac{1}{2}}$ is also continuous but has no finite derivative at x_0, showing a different failure mode.

Source lineage: Classic-method variant · differentiability_and_one_sided_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Proof Methods Open-text method adaptation

Q030 · Prove a Translated Absolute Value Is Not Differentiable at Its Corner

Problem

Let a be any real number. Prove that $f(x) = |x - a|$ is continuous at $x = a$, but has left derivative -1 and right derivative 1 there, and hence is not differentiable.

Answer

f is continuous at a, with $f'_-(a) = -1$ and $f'_+(a) = 1$, so $f'(a)$ does not exist.

Knowledge points

- translation preserves the corner of an absolute-value graph
- one-sided difference quotients
- continuity versus differentiability

Reading and method choice

Use $h = x - a$ as the increment from the base point a. The local problem then becomes exactly the behavior of $|h|$ near 0.

Detailed derivation

1. Since $f(a) = |a - a| = 0$ and $|x - a| \to 0$ as $x \to a$, $\lim_{x\to a} f(x) = f(a)$, so f is continuous at a.
2. The difference quotient is $\frac{f(a+h)-f(a)}{h} = \frac{|h|}{h}$.
3. For $h \to 0^-, |h| = -h$, so the quotient is -1 and $f'_-(a) = -1$.
4. For $h \to 0^+, |h| = h$, so the quotient is 1 and $f'_+(a) = 1$. The values differ, so $f'(a)$ does not exist.

Common pitfalls

- assuming the base point a must be 0
- omitting $f(a)$ in the continuity check
- thinking that translation removes the corner

Verification

With $u = x - a$, the graph is simply $y = |u|$ shifted so its vertex is (a,0); the one-sided slopes remain -1 and 1.

Method takeaway

For a translated local property, the increment $h = x - a$ moves the base point back to 0.

Extension

For $c|x - a| + d$ with $c \neq 0$, differentiability still fails at $x = a$; the one-sided derivatives are $-c$ and c.

Source lineage: Open-text method adaptation · differentiability_and_one_sided_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q031 · Product Rule and Simplification

Problem

Differentiate $y = (x^2 + 1)(x^3 - 2x)$, and simplify the result to a polynomial.

Answer

$y' = 5x^4 - 3x^2 - 2$

Knowledge points

- product rule
- power rule
- combining like terms

Reading and method choice

Name the two factors u and v and use $(uv)' = u'v + uv'$. Expanding afterward provides a useful check against a missing term.

Detailed derivation

1. Let $u = x^2 + 1$ and $v = x^3 - 2x$. Then $u' = 2x$ and $v' = 3x^2 - 2$.
2. By the product rule, $y' = 2x(x^3 - 2x) + (x^2 + 1)(3x^2 - 2)$.
3. The two expanded expressions are $2x^4 - 4x^2$ and $3x^4 + x^2 - 2$.
4. Combining like terms gives $y' = (2x^4 + 3x^4) + (-4x^2 + x^2) - 2 = 5x^4 - 3x^2 - 2$.

Common pitfalls

- writing $u'v'$
- differentiating only one factor
- losing the constant term -2 when expanding $(x^2 + 1)(3x^2 - 2)$

Verification

Expanding first gives $y = x^5 - x^3 - 2x$, whose termwise derivative is also $5x^4 - 3x^2 - 2$.

Method takeaway

A product derivative retains two cross terms; when expansion is possible, use it as an independent check.

Extension

For three factors, repeated use of the product rule gives a three-term sum in which one factor is differentiated at a time.

Source lineage: Classic-method variant · elementary_differentiation_rules · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.3. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Fill in the blank | Methods | Classic-method variant

Q032 · Identify the Inner and Outer Functions

Problem

If $y = \sin(3x^2 - 1)$, then $y' =$ ____.

Answer

$6x\cos(3x^2 - 1)$

Knowledge points

- chain rule
- $(\sin u)' = \cos u \cdot u'$
- polynomial differentiation

Reading and method choice

The outer function is $\sin u$ and the inner function is $u = 3x^2 - 1.$ Differentiate the outer function with respect to u and multiply by the derivative of the inner function.

Detailed derivation

1. Set $u = 3x^2 - 1$, so $y = \sin u$.
2. The outer derivative is $\frac{dy}{du} = \cos u$.
3. The inner derivative is $\frac{du}{dx} = 6x$.
4. By the chain rule, $y' = (\frac{dy}{du})(\frac{du}{dx}) = \cos(3x^2 - 1) \cdot 6x = 6x\cos(3x^2 - 1)$.

Common pitfalls

- writing only $\cos(3x^2 - 1)$ and omitting 6x
- using $-\sin u$ as the derivative of $\sin u$
- differentiating the inner constant -1 as -1

Verification

At $x = 0$ the inner rate is 0, and the formula gives $y'(0) = 0$, as a composite rate should include the inner factor.

Method takeaway

For the chain rule, differentiate the outer layer while keeping the inner expression, then multiply by the inner derivative.

Extension

For $y = \sin^3(3x^2 - 1)$, an additional outer power layer would also need the chain rule.

Source lineage: Classic-method variant · chain_rule_and_composites · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.6. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Multiple choice Methods Original synthesis / diagnosis

Q033 · Distinguish Four Differentiation Formulas

Problem

Assume all differentiability, domain, nonzero-denominator, and inverse-continuity conditions required by the formulas hold. For the inverse formula, let $y_0 = f(x_0)$ and $f'(x_0) \neq 0$. Which formulas are correct?

A. $(fg)' = f'g + fg'$

B. $(\frac{f}{g})' = \frac{f'g-fg'}{g^2}$ for $(g \neq 0)$

C. $(f \circ g)'(x) = f'(x)g'(x)$

D. $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}$

Answer

A, B, D

Knowledge points

- product rule
- quotient rule
- chain rule
- inverse-function derivative formula

Reading and method choice

A, B, and D are standard formulas. C is wrong because the derivative of the outer function must be evaluated at the inner value $g(x)$, not at x.

Detailed derivation

1. The product rule is $(fg)' = f'g + fg'$, so A is correct.
2. Where $g \neq 0$, the quotient rule is $(\frac{f}{g})' = \frac{f'g-fg'}{g^2}$, so B is correct.
3. The correct chain rule is $(f \circ g)'(x) = f'(g(x))g'(x)$, so C is generally false.
4. Under the stated inverse-function conditions, $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}$, so D is correct.

Common pitfalls

- writing the derivative of a product as the product of derivatives
- reversing the numerator in the quotient rule
- omitting the evaluation point $g(x)$ in the chain rule

Verification

Take $f(u) = u^2$ and $g(x) = x + 1$. The composite derivative is $2(x + 1)$, while C gives 2x, directly disproving C.

Method takeaway

A formula includes both its algebraic structure and its evaluation points; $f'(g(x))$ in the chain rule is essential.

Extension

Writing $\frac{f}{g}$ as $f \cdot g^{-1}$ and applying the product and chain rules also derives the quotient rule.

Source lineage: Original synthesis / diagnosis · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.7. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q034 · Quotient Rule for a Rational Function

Problem

Differentiate $y = \frac{x^2+1}{x-1}$, and state where the derivative formula applies.

Answer

$y' = \frac{x^2-2x-1}{(x-1)^2}, x \neq 1$

Knowledge points

- quotient rule
- nonzero denominator condition
- numerator simplification

Reading and method choice

Treat the numerator and denominator as u and v. Use the quotient rule in its ordered form and retain the restriction $x \neq 1$.

Detailed derivation

1. Let $u = x^2 + 1$ and $v = x - 1$. Then $u' = 2x$ and $v' = 1$.
2. By the quotient rule, $y' = \frac{2x(x-1)-(x^2+1)\cdot 1}{(x-1)^2}$.
3. Simplify the numerator: $2x^2 - 2x - x^2 - 1 = x^2 - 2x - 1$.
4. Thus $y' = \frac{x^2-2x-1}{(x-1)^2}$; both the original function and this derivative are considered only for $x \neq 1$.

Common pitfalls

- using $x - 1$ instead of $(x-1)^2$ in the denominator
- reversing the numerator terms and changing the overall sign
- omitting $x \neq 1$

Verification

Polynomial division gives $y = x + 1 + \frac{2}{x-1}$, whose derivative is $1 - \frac{2}{(x-1)^2}$; combining fractions reproduces $\frac{x^2-2x-1}{(x-1)^2}$.

Method takeaway

For a quotient, preserve both the numerator order and the original nonzero-denominator condition.

Extension

Compare the work required by algebraic division followed by differentiation with direct use of the quotient rule.

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Standard **Calculation** **Methods** **Classic-method variant**

Q035 · A Product Nested Inside a Quotient

Problem

For $x > 0$ and $x \neq 1$, differentiate $y = \frac{(x^2+1)\ln x}{x-1}$, and give a result over a common denominator.

Answer

$$y' = \frac{[(x^2-2x-1)\ln x] + \frac{(x^2+1)(x-1)}{x}}{(x-1)^2}.$$

Knowledge points

- Product rule
- Quotient rule
- Common-denominator algebra

Reading and method choice

First differentiate the numerator as a single function $N = (x^2+1)\ln x$, then apply the quotient rule to $\frac{N}{x-1}$. This organization prevents missing terms.

Detailed derivation

1. Let $N(x) = (x^2+1)\ln x$, so $y = \frac{N(x)}{x-1}$.
2. The product rule gives $N'(x) = 2x\ln x + \frac{x^2+1}{x}$.
3. The quotient rule gives $y' = \frac{N'(x)(x-1)-N(x)}{(x-1)^2}$.
4. Substitution yields $y' = \frac{[2x\ln x+\frac{x^2+1}{x}](x-1)-(x^2+1)\ln x}{(x-1)^2}$.
5. For the logarithmic terms, $2x(x-1)-(x^2+1) = x^2-2x-1$, which gives the stated answer.

Common pitfalls

- Omitting $\frac{x^2+1}{x}$ from the numerator derivative
- Reversing the quotient-rule numerator
- Using the formula at $x = 1$

Verification

Expanding the partially simplified form reproduces the final numerator term by term, and both forms have domain $x > 0, x \neq 1$.

Method takeaway

Name a complicated numerator first, then apply product and quotient rules layer by layer.

Extension

If the denominator is changed to $x - a$, replace each occurrence of $x - 1$ in the quotient-rule setup by $x - a$.

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Standard | Single choice | Methods | Classic-method variant

Q036 · A Composite of Arcsine and a Square Root

Problem

For $0 < x < 1$, which expression is the derivative of $y = \arcsin\sqrt{x}$?

A. $\frac{1}{\sqrt{1-x}}$

B. $\frac{1}{2\sqrt{x}\sqrt{1-x}}$

C. $\frac{1}{2\sqrt{x}(1-x)}$

D. $-\frac{1}{2\sqrt{x}\sqrt{1-x}}$

Answer

B. $\frac{1}{2\sqrt{x}\sqrt{1-x}}$.

Knowledge points

- Derivative of arcsine
- Derivative of a square root
- Chain rule

Reading and method choice

The outer function is $\arcsin u$ and the inner function is $u = \sqrt{x}$. The quantity $1 - u^2$ becomes $1 - x$, but the factor u' must still be included.

Detailed derivation

1. Set $u = \sqrt{x}$, so $y = \arcsin u$.
2. The outer derivative is $\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$.
3. The inner derivative is $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$.
4. By the chain rule, $y' = \frac{1}{2\sqrt{x}\sqrt{1-u^2}}$.
5. Since $u^2 = x$, $y' = \frac{1}{2\sqrt{x}\sqrt{1-x}}$, so the answer is B.

Common pitfalls

- Omitting the factor $\frac{1}{2\sqrt{x}}$
- Replacing $1-u^2$ by $1-\sqrt{x}$
- Adding an incorrect negative sign to the arcsine derivative

Verification

For $0 < x < 1$ both radicals are positive, so the derivative is positive, consistent with the function increasing with x.

Method takeaway

For an inverse-trigonometric composite, keep the inner variable until all chain-rule factors are present.

Extension

For $y = \arcsin\sqrt{ax+b}$, one must also multiply by the derivative a of $ax+b$.

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Standard | Multiple choice | Methods | Original synthesis / diagnosis

Q037 · Distinguishing Four Differentiation Rules

Problem

Assume the functions involved are differentiable at the relevant points and all expressions are defined. Which statements are correct?

A. $[u(v(x))]' = u'(v(x))v'(x)$

B. $[u(x)v(x)]' = u'(x)v'(x)$

C. $[\frac{u(x)}{v(x)}]' = \frac{u'(x)v(x)-u(x)v'(x)}{v^2(x)}$, where $v(x) \neq 0$

D. If f has a differentiable inverse and $f'(f^{-1}(y)) \neq 0$, then $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$

Answer

A, C, and D.

Knowledge points

- Chain rule
- Product and quotient rules
- Inverse-function derivative formula

Reading and method choice

Check both the structure and the conditions of each formula. Statement B incorrectly replaces the product rule by a product of derivatives; the other three are standard formulas.

Detailed derivation

1. A is the chain rule: differentiate the outer function at $v(x)$, then multiply by the inner derivative, so A is correct.
2. B is false; the correct product rule is $(uv)' = u'v + uv'$.
3. C has the correct numerator order, squared denominator, and condition $v(x) \neq 0$, so C is correct.
4. D is the correct inverse derivative formula when the original derivative at the corresponding point is nonzero.
5. Therefore the correct selections are A, C, and D.

Common pitfalls

- Remembering the product rule as $u'v'$
- Omitting the squared denominator in the quotient rule

- Omitting the corresponding point $f^{-1}(y)$ in the inverse formula

Verification

Taking $u = v = x$ immediately disproves $B:$ the left side is $(x^2)' = 2x$, while the proposed right side is 1.

Method takeaway

A differentiation formula is safest when its structure and its conditions are remembered together.

Extension

Derive the quotient rule from the product rule and $(\frac{1}{v})' = -\frac{v'}{v^2}$.

Source lineage: Original synthesis / diagnosis · elementary_differentiation_rules · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.3. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q038 · A Trigonometric Square Inside a Logarithm

Problem

Differentiate $y = \ln[1 + \sin^2(3x)]$, listing the layers from outside to inside.

Answer

$y' = \frac{6\sin(3x)\cos(3x)}{1+\sin^2(3x)} = \frac{3\sin(6x)}{1+\sin^2(3x)}$.

Knowledge points

- Multilevel chain rule
- Logarithmic composite
- Trigonometric identity

Reading and method choice

There are four layers: ln, addition and squaring, sine, and 3x. Differentiate from the outside inward, multiplying the derivative contributed by every layer.

Detailed derivation

1. Set $u = 1 + \sin^2(3x)$, so $y = \ln u$ and $y' = \frac{u'}{u}$.
2. Set $v = \sin(3x)$, so $u = 1 + v^2$ and $u' = 2vv'$.
3. The chain rule gives $v' = \cos(3x) \cdot 3$.
4. Combining the factors gives $y' = \frac{6\sin(3x)\cos(3x)}{1+\sin^2(3x)}$.
5. Using $2\sin A\cos A = \sin 2A$, this may also be written as $\frac{3\sin(6x)}{1+\sin^2(3x)}$.

Common pitfalls

- Missing the innermost factor 3
- Writing the derivative of $\sin^2(3x)$ as $2\sin(3x)$
- Misapplying the derivative of ln to the whole expression

Verification

At $x = 0$, the first-order change of the original function is zero, and the derived formula also evaluates to zero.

Method takeaway

Naming each layer of a composite function is a reliable way to retain every chain-rule factor.

Extension

If 3x is replaced by a differentiable $\varphi(x)$, replace the last factor 3 by $\varphi'(x)$.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.5, openstax-calculus-v1-3.9. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Calculation** **Methods** **Classic-method variant**

Q039 · Both Base and Exponent Depend on x

Problem

Differentiate $y = (x^2 + 1)^{\sin x}$ using logarithmic differentiation, and state its domain.

Answer

$y' = (x^2 + 1)^{\sin x}[\cos x \cdot \ln(x^2 + 1) + \frac{2x \sin x}{x^2+1}]$, **with domain all real x.**

Knowledge points

- Logarithmic differentiation
- Variable powers
- Chain rule

Reading and method choice

The base $x^2 + 1$ is always positive, so taking natural logarithms converts the variable exponent into a product that can be differentiated normally.

Detailed derivation

1. Since $x^2 + 1 > 0$ for every real $x, y > 0$ and the original function is defined on $\mathbb{R}$.
2. Take logarithms: $\ln y = \sin x \cdot \ln(x^2 + 1)$.
3. Differentiate: $\frac{y'}{y} = \cos x \cdot \ln(x^2 + 1) + \frac{\sin x \cdot 2x}{x^2+1}$.
4. Multiply by y to obtain $y' = y[\cos x \cdot \ln(x^2 + 1) + \frac{2x \sin x}{x^2+1}]$.
5. Substitute $y = (x^2 + 1)^{\sin x}$ to obtain the stated answer.

Common pitfalls

- Applying the ordinary power rule directly
- Changing the product into a sum after taking logarithms
- Forgetting to multiply back by y

Verification

At $x = 0, y = 1$ and both terms in brackets vanish, so $y'(0) = 0$, consistent with the local structure of the original function.

Method takeaway

For a variable power with positive base, logarithms move the variable exponent into an ordinary product.

Extension

For $y = u(x)^{v(x)}$ with $u > 0, y' = y[v' \ln u + \frac{vu'}{u}]$.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.9. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Proof Methods Classic-method variant

Q040 · Proving the Inverse Derivative Formula from the Definition

Problem

Suppose f is differentiable at x_0 with $f'(x_0) \neq 0$ and is one-to-one near x_0. Its inverse g is continuous at $y_0 = f(x_0)$. Prove from the derivative definition that $g'(y_0) = \frac{1}{f'(x_0)}$.

Answer

Set $x = g(y)$, rewrite $\frac{g(y)-g(y_0)}{y-y_0}$ as $\frac{x-x_0}{f(x)-f(x_0)}$, and take the limit to obtain $g'(y_0) = \frac{1}{f'(x_0)}$.

Knowledge points

- Inverse correspondence
- Derivative definition
- Reciprocal limits

Reading and method choice

The key is to rewrite the inverse difference quotient as the reciprocal of the original difference quotient. Continuity of g at y_0 ensures that $x = g(y)$ approaches x_0 when y approaches y_0.

Detailed derivation

1. From $y_0 = f(x_0)$ and $g = f^{-1}$, we have $g(y_0) = x_0$.
2. For $y \neq y_0$, set $x = g(y)$. One-to-one correspondence gives $x \neq x_0$ and $y = f(x)$.
3. Rewrite the inverse difference quotient as $\frac{g(y)-g(y_0)}{y-y_0} = \frac{x-x_0}{f(x)-f(x_0)}$.
4. Because g is continuous at $y_0, y \to y_0$ implies $x = g(y) \to x_0$; the right side is the reciprocal of the original difference quotient.
5. That original quotient tends to $f'(x_0) \neq 0$, so its reciprocal tends to $\frac{1}{f'(x_0)}$.
6. By the derivative definition, $g'(y_0) = \frac{1}{f'(x_0)}$, as required.

Common pitfalls

- Failing to justify that $y \to y_0$ implies $x \to x_0$
- Ignoring the nonzero condition before taking a reciprocal
- Confusing the inputs x_0 and y_0

Verification

For $f(x) = 2x + 3$, $f' = 2$ and the inverse $g(y) = \frac{y-3}{2}$ indeed has derivative $\frac{1}{2}$.

Method takeaway

The inverse derivative formula comes directly from reciprocal ratios of corresponding increments.

Extension

If $f'(x_0) = 0$, the reciprocal quotient may have no finite limit, so the inverse must be examined from its definition.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.7. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Fill in the blank** **Methods** **Classic-method variant**

Q041 · Determining Two Parameters in a Piecewise Function

Problem

Let $f(x) = \begin{cases} ax+b, & x<1, \\ x^2+1, & x \geq 1. \end{cases}$ For f to be differentiable at $x=1$, $a=$ ___ and $b=$ ___.

Answer

$a=2$ **and** $b=0$.

Knowledge points

- Differentiability of piecewise functions
- Continuity as a necessary condition
- One-sided derivatives

Reading and method choice

First connect the two function values by continuity, then equate the left and right derivatives. These two independent conditions determine a and b.

Detailed derivation

1. The right-hand piece gives $f(1)=1^2+1=2$.
2. The left-hand limit is $a+b$, so continuity requires $a+b=2$.
3. The left derivative is a, while the derivative of x^2+1 at $x=1$ is 2.
4. Differentiability requires equal one-sided derivatives, hence $a=2$.
5. Substituting into $a+b=2$ gives $b=0$; both conditions then hold.

Common pitfalls

- Equating derivatives without first checking continuity
- Computing the right derivative as 1
- Omitting b from the left limit

Verification

After substitution, the pieces are 2x and $x^2 + 1$; both have value 2 and derivative 2 at $x = 1$.

Method takeaway

At a joining point, differentiability requires both matching values and matching slopes.

Extension

If only continuity at $x = 1$ is required, the single condition $a + b = 2$ leaves infinitely many choices.

Source lineage: Classic-method variant · elementary_differentiation_rules · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.3. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q042 · Second and Third Derivatives of an Exponential-Cosine Product

Problem

Let $y = e^{2x} \cos x$. Find y'' and y'''.

Answer

$y'' = e^{2x}(3 \cos x - 4 \sin x)$**, and** $y''' = e^{2x}(2 \cos x - 11 \sin x)$**.**

Knowledge points

- Successive differentiation
- Product rule
- Exponential and trigonometric functions

Reading and method choice

After every differentiation, the expression remains e^{2x} times a linear combination of sine and cosine. Preserve this structure and update the coefficients.

Detailed derivation

1. First differentiate: $y' = e^{2x}(2 \cos x - \sin x)$.
2. Apply the product rule again: $y'' = e^{2x}[2(2 \cos x - \sin x) + (-2 \sin x - \cos x)]$.
3. Simplification gives $y'' = e^{2x}(3 \cos x - 4 \sin x)$.
4. Differentiate once more: $y''' = e^{2x}[2(3 \cos x - 4 \sin x) + (-3 \sin x - 4 \cos x)]$.
5. Combining coefficients gives $y''' = e^{2x}(2 \cos x - 11 \sin x)$.

Common pitfalls

- Missing the factor 2 from $(e^{2x})'$ at each stage
- Using the wrong sign for the derivative of sine
- Combining coefficients before completing the product rule

Verification

At $x = 0$, the formulas give $y''(0) = 3$ and $y'''(0) = 2$; the unsimplified stage-by-stage expressions give the same values.

Method takeaway

Products of an exponential with a sine-cosine combination stay in the same function family under differentiation.

Extension

The coefficient update for $e^{2x}(A\cos x + B\sin x)$ can be recorded as $(A, B) \mapsto (2A + B, -A + 2B)$.

Source lineage: Classic-method variant · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | True/false with justification | Methods | Original synthesis / diagnosis

Q043 · Does the nth Derivative of a Product Have Only Two Terms?

Problem

True or false, with justification: For every $n \geq 2$, if u and v have the needed derivatives, then $(uv)^{(n)} = u^{(n)}v + uv^{(n)}$.

Answer

False. There are intermediate terms in which derivatives are distributed between both factors.

Knowledge points

- Higher derivatives of a product
- Second-order product rule
- Counterexamples

Reading and method choice

It is enough to inspect $n = 2$. Differentiating the product rule once more produces two identical $u'v'$ terms.

Detailed derivation

1. The first-order product rule is $(uv)' = u'v + uv'$.
2. Differentiating again gives $(uv)'' = u''v + u'v' + u'v' + uv''$.
3. Thus $(uv)'' = u''v + 2u'v' + uv''$, so the proposed formula misses a middle term.
4. Take $u = v = x^2$; then $(uv)'' = (x^4)'' = 12x^2$.
5. The proposed right side is $2x^2 + 2x^2 = 4x^2$, so the statement is false.

Common pitfalls

- Extending the first-order product rule to only two endpoint terms
- Missing the repeated $u'v'$ term
- Failing to test the smallest case $n = 2$

Verification

The middle terms vanish when one factor is constant, but that special case does not establish the general formula.

Method takeaway

A higher derivative of a product must account for every distribution of the n differentiations between the two factors.

Extension

In the complete nth-order formula, the coefficients are the binomial coefficients $C(n, k)$.

Source lineage: Original synthesis / diagnosis · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q044 · Higher Derivatives of an Exponential Plus a Sine

Problem

Let $y = \sin(2x) + e^{3x}$. Find y'' and y'''.

Answer

$y'' = -4\sin(2x) + 9e^{3x}$, and $y''' = -8\cos(2x) + 27e^{3x}$.

Knowledge points

- Successive differentiation
- Trigonometric composites
- Exponential composites

Reading and method choice

The two terms can be differentiated separately. Each differentiation of $\sin(2x)$ contributes a factor 2, while each differentiation of e^{3x} contributes a factor 3.

Detailed derivation

1. The first derivative is $y' = 2\cos(2x) + 3e^{3x}$.
2. Differentiate again: $(2\cos(2x))' = -4\sin(2x)$, and $(3e^{3x})' = 9e^{3x}$.
3. Hence $y'' = -4\sin(2x) + 9e^{3x}$.
4. Differentiate the second derivative: $[-4\sin(2x)]' = -8\cos(2x)$, and $(9e^{3x})' = 27e^{3x}$.
5. Therefore $y''' = -8\cos(2x) + 27e^{3x}$.

Common pitfalls

- Multiplying by the inner coefficient only once overall
- Dropping the negative sign from differentiating cosine
- Leaving the derivative of $3e^{3x}$ as $3e^{3x}$

Verification

At $x = 0$, $y''(0) = 9$ and $y'''(0) = -8 + 27 = 19$, matching the unsimplified stage-by-stage expressions.

Method takeaway

Higher derivatives of a linear combination are taken termwise, and each order repeats the inner chain factor.

Extension

In general, $[\sin(ax)]^{(n)} = a^n \sin(ax + \frac{n\pi}{2})$ and $(e^{bx})^{(n)} = b^n e^{bx}$.

Source lineage: Classic-method variant · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Proof | Methods | Classic-method variant

Q045 · The nth Derivative of the Reciprocal of a Linear Factor

Problem

For every integer $n \geq 0$, prove by induction that if $y = \frac{1}{x-a}, x \neq a$, then $y^{(n)} = \frac{(-1)^n n!}{(x-a)^{n+1}}$.

Answer

The formula holds; differentiating the nth-order formula once produces the formula of order $n+1$.

Knowledge points

- nth derivatives
- Mathematical induction
- Differentiating negative integer powers

Reading and method choice

Verify the base case $n = 0$, then assume the result for $n = k$. Differentiating the assumed identity combines $k!$ with the exponent $k+1$ to form $(k+1)!$.

Detailed derivation

1. For $n = 0$, the right side is $\frac{(-1)^0 0!}{x-a} = \frac{1}{x-a} = y$, so the base case holds.
2. Assume that $y^{(k)} = (-1)^k k!(x-a)^{-k-1}$.
3. Differentiate this identity: $y^{(k+1)} = (-1)^k k! \cdot [-(k+1)](x-a)^{-k-2}$.
4. Combining signs and factorials gives $y^{(k+1)} = \frac{(-1)^{k+1}(k+1)!}{(x-a)^{k+2}}$.
5. This is exactly the claimed formula with n replaced by $k+1$, so the induction step holds.
6. Therefore the formula is valid for every $n \geq 0$.

Common pitfalls

- Starting at $n = 1$ even though $n = 0$ is included
- Failing to decrease the power by one
- Not simplifying $(k+1)k!$ to $(k+1)!$

Verification

For $n = 1$ and $n = 2$, the formula gives $-\frac{1}{(x-a)^2}$ and $\frac{2}{(x-a)^3}$, matching direct differentiation.

Method takeaway

For a higher-derivative formula, the induction step is often just one more differentiation.

Extension

If $x - a$ is replaced by $bx + c$, each differentiation contributes an additional factor b.

Source lineage: Classic-method variant · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q046 · First Derivative of an Exponential Implicit Curve

Problem

The curve $e^{xy}+x+y=1$ passes through $(0,0)$. Find the general expression for y' and the tangent slope at that point.

Answer

$y'=-\frac{ye^{xy}+1}{xe^{xy}+1}$; **at** $(0,0)$, $y'=-1$**.**

Knowledge points

- Chain rule for e^{xy}
- Product rule for xy
- Implicit tangent slope

Reading and method choice

The derivative of the inner product xy is $y+xy'$, and the outer exponential contributes e^{xy}. Then collect y'.

Detailed derivation

1. Differentiate e^{xy} to obtain $e^{xy}(y+xy')$.
2. Differentiate $x+y$ to obtain $1+y'$; the derivative of the constant 1 is 0.
3. Thus $e^{xy}(y+xy')+1+y'=0$.
4. Collecting y' gives $(xe^{xy}+1)y'=-(ye^{xy}+1)$.
5. Therefore $y'=-\frac{ye^{xy}+1}{xe^{xy}+1}$, and at $(0,0)$ it equals -1.

Common pitfalls

- Writing $\frac{d(e^{xy})}{dx}$ as $e^{xy}xy'$
- Omitting the y term from $\frac{d(xy)}{dx}$
- Treating e^0 as 0

Verification

$(0,0)$ satisfies $e^0+0+0=1$. Substituting $y'=-1$ into the differentiated relation gives $0+1-1=0$.

Method takeaway

For an implicit composite, apply the chain rule from outside inward and differentiate any inner product completely.

Extension

The tangent is $y = -x$, while the normal slope is 1.

Source lineage: Classic-method variant · implicit_differentiation · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.8. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Calculation | Methods | Original synthesis / diagnosis

Q047 · Differentiating a Trigonometric Implicit Curve

Problem

The curve $\sin(x+y)=xy$ passes through $(0,0)$. Find the general expression for y' and the tangent line at that point.

Answer

$y' = \frac{y-\cos(x+y)}{\cos(x+y)-x}$**; at** $(0,0)$, $y' = -1$ **and the tangent is** $y=-x$**.**

Knowledge points

- Chain rule for $\sin(x+y)$
- Product rule for xy
- Implicit tangent line

Reading and method choice

Both sides contain $y(x)$. Differentiate the outer sine on the left and use the product rule on xy.

Detailed derivation

1. Differentiate the left side to obtain $\cos(x+y)(1+y')$.
2. Differentiate xy to obtain $y+xy'$.
3. Expand: $\cos(x+y)+\cos(x+y)y' = y+xy'$.
4. Collect y' to get $[\cos(x+y)-x]y' = y-\cos(x+y)$.
5. Hence $y' = \frac{y-\cos(x+y)}{\cos(x+y)-x}$; at $(0,0)$, $y' = -1$ and the tangent is $y=-x$.

Common pitfalls

- Omitting the factor $1+y'$
- Writing $\frac{d(xy)}{dx}$ as xy'
- Reversing the sign of $\cos(x+y)-x$

Verification

$(0,0)$ satisfies $\sin 0 = 0$, and the differentiated equation there is $1\cdot(1+y')=0$, so $y'=-1$.

Method takeaway

Differentiate every y-containing expression on both sides; none of it is constant with respect to x.

Extension

If $\cos(x + y) - x = 0$, the explicit fraction fails and local vertical behavior must be checked.

Source lineage: Original synthesis / diagnosis · implicit_differentiation · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.8. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Calculation | Methods | Classic-method variant

Q048 · First Derivative and Tangent of a Parametric Curve

Problem

For $x = t^2 + 1$ and $y = t^3 - t$, find $\frac{dy}{dx}$ at $t = 1$ and write the tangent line.

Answer

$\frac{dy}{dx} = 1$; the point is $(2, 0)$, and the tangent is $y = x - 2$.

Knowledge points

- First derivative of a parametric curve
- Coordinates from a parameter
- Point-slope form

Reading and method choice

Differentiate x and y with respect to t, then divide when $\frac{dx}{dt} \neq 0$. The tangent line also requires the point corresponding to $t = 1$.

Detailed derivation

1. $\frac{dx}{dt} = 2t$ and $\frac{dy}{dt} = 3t^2 - 1$.
2. At $t = 1$, $\frac{dx}{dt} = 2 \neq 0$ and $\frac{dy}{dt} = 2$.
3. Thus $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2}{2} = 1$.
4. The parameter point is $x = 1^2 + 1 = 2$ and $y = 1^3 - 1 = 0$.
5. Point-slope form gives $y - 0 = 1(x - 2)$, or $y = x - 2$.

Common pitfalls

- Reversing the ratio for $\frac{dy}{dx}$
- Finding the slope but not the parameter point
- Failing to check $\frac{dx}{dt} \neq 0$

Verification

Near $t = 1, dx \approx 2dt$ and $dy \approx 2dt$, so $\frac{\Delta y}{\Delta x} \approx 1$.

Method takeaway

A parametric slope is the ratio of the two rates with respect to the parameter, provided $\frac{dx}{dt}$ is nonzero.

Extension

At $t = 0$, $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = -1$, so the curve has a vertical tangent.

Source lineage: Classic-method variant · parametric_differentiation · reference IDs: openstax-calculus-v2-7.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Fill in the blank | Methods | Classic-method variant

Q049 · The Parametric Derivative Formula and Its Value

Problem

If $x = t^2$ and $y = t^4 + t$, then when $\frac{dx}{dt} \neq 0$, $\frac{dy}{dx} =$ ____; at $t = 1$ its value is ____.

Answer

$\frac{\frac{dy}{dt}}{\frac{dx}{dt}}$; $\frac{5}{2}$.

Knowledge points

- Parametric derivative formula
- Power rule
- Evaluation at a parameter value

Reading and method choice

State the general formula first, then differentiate x and y separately. At $t = 1$, $\frac{dx}{dt}$ is nonzero, so division is valid.

Detailed derivation

1. The first parametric derivative is $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$.
2. Here $\frac{dx}{dt} = 2t$.
3. Also $\frac{dy}{dt} = 4t^3 + 1$.
4. At $t = 1$, $\frac{dx}{dt} = 2$ and $\frac{dy}{dt} = 5$.
5. Thus $\frac{dy}{dx} = \frac{5}{2}$, so the blanks are $\frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ and $\frac{5}{2}$.

Common pitfalls

- Multiplying the two derivatives
- Differentiating t^4 as 4t
- Ignoring the zero denominator at $t = 0$

Verification

At $t = 1, x$ changes at rate 2 and y at rate 5, so the y-change per unit x-change is $\frac{5}{2}$.

Method takeaway

A parametric derivative is the ratio of two rates measured with respect to the same parameter.

Extension

At $t = 0$ the denominator vanishes, so the tangent direction must be analyzed separately.

Source lineage: Classic-method variant · parametric_differentiation · reference IDs: openstax-calculus-v2-7.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Single choice Methods Original synthesis / diagnosis

Q050 · Condition for a Vertical Tangent of a Parametric Curve

Problem

Suppose the parameter derivatives exist at $t = t_0$. Which condition most directly indicates a vertical tangent?

A. $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} \neq 0$

B. $\frac{dx}{dt} \neq 0$ and $\frac{dy}{dt} = 0$

C. $\frac{dx}{dt} = \frac{dy}{dt} = 1$

D. $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$, with no further analysis needed

Answer

A

Knowledge points

- Tangent direction of a parametric curve
- Vertical tangent
- Zero velocity component

Reading and method choice

The velocity vector is $(\frac{dx}{dt}, \frac{dy}{dt})$. A zero horizontal component and nonzero vertical component give a vertical direction.

Detailed derivation

1. The instantaneous direction of a parametric curve is determined by $(\frac{dx}{dt}, \frac{dy}{dt})$.
2. If $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} \neq 0$, this vector is parallel to the y-axis, so the tangent is vertical.
3. Condition B gives a horizontal tangent.
4. Condition C gives the finite slope $\frac{\frac{dy}{dt}}{\frac{dx}{dt}} = 1$.
5. In D the velocity vector is zero, so first derivatives alone are inconclusive; therefore A is correct.

Common pitfalls

- Reversing horizontal and vertical conditions
- Assuming $\frac{0}{0}$ automatically means vertical
- Omitting the condition $\frac{dy}{dt} \neq 0$

Verification

For $x = t^2, y = t$ at $t = 0, \frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 1$. Eliminating t gives $x = y^2$, whose tangent at the origin is $x = 0$.

Method takeaway

Read a parametric tangent from the velocity vector rather than treating zero as an ordinary denominator.

Extension

If both parameter derivatives vanish, lower-order nonzero changes must be studied; no automatic conclusion is available.

Source lineage: Original synthesis / diagnosis · parametric_differentiation · reference IDs: openstax-calculus-v2-7.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Calculation** **Methods** **Open-text method adaptation**

Q051 · Volume Rate of an Expanding Sphere

Problem

The radius of a spherical balloon increases at $2\,\mathrm{cm\,s^{-1}}$. How fast is its volume increasing when the radius is $5\,\mathrm{cm}$?

Answer

$\frac{dV}{dt} = 200\pi\,\mathrm{cm^3\,s^{-1}}$.

Knowledge points

- Volume of a sphere
- Chain rule
- Units in related rates

Reading and method choice

Let $r = r(t)$ and $V = (\frac{4}{3})\pi r^3$. Differentiate with respect to t before inserting r and $\frac{dr}{dt}$.

Detailed derivation

1. Use cm for r and s for t; the given rate is $\frac{dr}{dt} = 2\,\mathrm{cm\,s^{-1}}$.
2. The sphere volume is $V = (\frac{4}{3})\pi r^3$.
3. Differentiate in time: $\frac{dV}{dt} = \frac{4\pi r^2 \cdot dr}{dt}$.
4. Substitute $r = 5$ and $\frac{dr}{dt} = 2$ to obtain $\frac{dV}{dt} = 4\pi \cdot 25 \cdot 2$.
5. Thus $\frac{dV}{dt} = 200\pi\,\mathrm{cm^3\,s^{-1}}$; the positive sign indicates increasing volume.

Common pitfalls

- Omitting $\frac{dr}{dt}$
- Treating 5 as the diameter
- Using $\frac{cm}{s}$ for a volume rate

Verification

The dimension is $\mathrm{cm}^2 \cdot \mathrm{cm\,s}^{-1} = \mathrm{cm}^3\,\mathrm{s}^{-1}$, and the positive sign agrees with expansion.

Method takeaway

When differentiating a geometric formula in time, attach the appropriate time derivative to every changing length.

Extension

If the radius decreases at the same speed, $\frac{dr}{dt} = -2\,\mathrm{cm\,s}^{-1}$ and $\frac{dV}{dt} = -200\pi\,\mathrm{cm}^3\,\mathrm{s}^{-1}$.

Source lineage: Open-text method adaptation · related_rates · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.1. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | True/false with justification | Methods | Classic-method variant

Q052 · Differentiability and Existence of a Derivative in One Variable

Problem

True or false, with justification: For a one-variable function $y = f(x)$, differentiability at a point is equivalent to existence of the derivative there, and the differential coefficient is $f'(x)$.

Answer

True. Differentiability means $\Delta y = A\Delta x + o(\Delta x)$; division by Δx yields a derivative equal to A, and conversely a derivative gives the same decomposition.

Knowledge points

- Definition of differentiability
- Definition of derivative
- Equivalence in one variable
- Differential coefficient

Reading and method choice

The bridge is the increment decomposition $\Delta y = A\Delta x + o(\Delta x)$. Dividing by Δx produces the difference-quotient limit.

Detailed derivation

1. If f is differentiable at x, there is a constant A with $\Delta y = A\Delta x + o(\Delta x)$.
2. For $\Delta x \neq 0$, divide by Δx : $\frac{\Delta y}{\Delta x} = A + \frac{o(\Delta x)}{\Delta x}$.
3. As $\Delta x \to 0$, the second term tends to 0, so $f'(x)$ exists and equals A.
4. Conversely, if $f'(x)$ exists, then $\frac{\Delta y}{\Delta x} - f'(x) \to 0$.
5. Let $\alpha(\Delta x) = \Delta y - f'(x)\Delta x$. Then $\frac{\alpha(\Delta x)}{\Delta x} \to 0$, so $\Delta y = f'(x)\Delta x + o(\Delta x)$.
6. Thus the statement is true and $dy = f'(x)dx$.

Common pitfalls

- Transferring multivariable subtleties to the one-variable case
- Citing the result without the increment decomposition
- Interpreting differentiable as numerically small

Verification

Both directions follow from the same difference-quotient limit, and A is uniquely determined as $f'(x)$.

Method takeaway

For one variable, existence of a derivative and differentiability are equivalent; the differential coefficient is the derivative.

Extension

Continuity alone is insufficient: $|x|$ is continuous at 0 but not differentiable there.

Source lineage: Classic-method variant · differentials_and_linear_approximation · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Calculation** **Methods** **Classic-method variant**

Q053 · Differential of a Logarithmic Composite

Problem

Let $y = \ln(1 + x^2)$. Find dy, and evaluate it at $x = 1$ with $dx = 0.02$.

Answer

$dy = [\frac{2x}{1+x^2}]dx$; **at** $x = 1$ **and** $dx = 0.02, dy = 0.02$.

Knowledge points

- Differential of a composite
- Derivative of logarithm
- Numerical differential

Reading and method choice

Apply the chain rule and then multiply by dx. At $x = 1$, the differential coefficient equals 1.

Detailed derivation

1. Let $u = 1 + x^2$, so $y = \ln u$.
2. Then $dy = (\frac{1}{u})du$ and $du = 2xdx$.
3. Therefore $dy = [\frac{2x}{1+x^2}]dx$.
4. At $x = 1$, the coefficient is $\frac{2}{1+1} = 1$.
5. With $dx = 0.02, dy = 1 \cdot 0.02 = 0.02$.

Common pitfalls

- Writing $d \ln u$ as $\ln u \cdot du$
- Omitting the factor 2x
- Evaluating the coefficient at $x + dx$ instead of the base point

Verification

The exact increment $\ln(1 + 1.02^2) - \ln 2 \approx 0.019998$, close to $dy = 0.02$.

Method takeaway

The form-invariance of differentials lets us work layer by layer: $dy = (\frac{1}{u})du$.

Extension

If $dx = -0.02$, then $dy = -0.02$, indicating a local decrease.

Source lineage: Classic-method variant · differentials_and_linear_approximation · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Calculation | Methods | Open-text method adaptation

Q054 · Approximating a Square Root with Differentials

Problem

Using only differential linearization near $x_0 = 4$, approximate $\sqrt{4.04}$.

Answer

$\sqrt{4.04} \approx 2.01$.

Knowledge points

- Linearization formula
- Derivative of square root
- Choice of base point
- Approximation symbol

Reading and method choice

Choose the convenient base point 4 and use $\Delta x = 0.04$ in $f(x_0 + \Delta x) \approx f(x_0) + f'(x_0)\Delta x$.

Detailed derivation

1. Let $f(x) = \sqrt{x}$, choose $x_0 = 4$, and set $\Delta x = 4.04 - 4 = 0.04$.
2. $f(4) = 2$ and $f'(x) = \frac{1}{2\sqrt{x}}$.
3. At the base point, $f'(4) = \frac{1}{4}$.
4. The differential is $dy = f'(4)\Delta x = (\frac{1}{4})(0.04) = 0.01$.
5. Hence $\sqrt{4.04} \approx f(4) + dy = 2 + 0.01 = 2.01$.

Common pitfalls

- Choosing 4.04 as the base point
- Using $f'(4) = \frac{1}{2}$
- Using an equality sign for an approximation

Verification

$2.01^2 = 4.0401$, only 0.0001 from 4.04. This supports the approximation but is not a rigorous error bound from linearization alone.

Method takeaway

Choose a nearby base point where both the function and derivative are easy to evaluate.

Extension

The same method can approximate $\sqrt{9.06}$ from $x_0 = 9$.

Source lineage: Open-text method adaptation · differentials_and_linear_approximation · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Multiple choice | Methods | Original synthesis / diagnosis

Q055 · Correct Statements about Differential Estimates

Problem

Let $y = f(x)$ be differentiable at x with $y \neq 0$. Which statements are correct? Select all that apply.

A. As $\Delta x \to 0, \Delta y = dy + o(\Delta x)$

B. dy is linear in dx

C. $\Delta y = dy$ for every finite dx

D. $\frac{dy}{y}$ is a first-order estimate of the relative increment $\frac{\Delta y}{y}$

Answer

A, B, and D

Knowledge points

- Differentiable increment formula
- Linearity of the differential
- Relative differential
- Limits of approximation

Reading and method choice

Differentiability says that the nonlinear remainder is negligible relative to a small Δx; it does not make every finite increment exactly linear.

Detailed derivation

1. A is true from $\Delta y = f'(x)\Delta x + o(\Delta x)$ and $dy = f'(x)dx$.
2. B is true because, at a fixed base point, $dy = f'(x)dx$ is linear in dx.
3. C is false for nonlinear functions; for x^2 an extra term $(\Delta x)^2$ remains.
4. D is true because, when $y \neq 0$, $\frac{\Delta y}{y} \approx \frac{dy}{y}$ gives a first-order relative-change estimate.
5. Therefore A, B, and D are correct.

Common pitfalls

- Treating a first-order approximation as an identity
- Ignoring $y \neq 0$ for a relative differential
- Treating $o(\Delta x)$ as a fixed constant

Verification

For $y = x^2, \Delta y = 2x\Delta x + (\Delta x)^2$ and $dy = 2x\Delta x$; this confirms A and B while disproving C.

Method takeaway

The differential retains first-order linear information; the relative differential normalizes it by the current function value.

Extension

If f is linear, then C also holds because the remainder is identically zero.

Source lineage: Original synthesis / diagnosis · differentials_and_linear_approximation · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Calculation | Methods | Original synthesis / diagnosis

Q056 · Differentiability of an Oscillatory Function at the Origin

Problem

Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$ Decide whether f is differentiable at 0, and find the complete derivative $f'(x)$.

Answer

f is differentiable at 0 with $f'(0) = 0$; for $x \neq 0$, $f'(x) = 2x\sin(\frac{1}{x}) - \cos(\frac{1}{x})$

Knowledge points

- using the definition at a special point
- bounded factor times an infinitesimal
- product and chain rules

Reading and method choice

The formula for $x \neq 0$ cannot simply be used at the origin; first principles are required there. At nonzero points, use the product and chain rules.

Detailed derivation

1. At the origin the difference quotient is $\frac{f(h)-f(0)}{h} = h\sin(\frac{1}{h})$ for $h \neq 0$.
2. Since $|h\sin(\frac{1}{h})| \leq |h| \to 0$, the quotient has a limit and $f'(0) = 0$.
3. For $x \neq 0$, applying the product rule to $x^2\sin(\frac{1}{x})$ gives the first term $2x\sin(\frac{1}{x})$.
4. The chain-rule contribution is $x^2 \cdot \cos(\frac{1}{x}) \cdot (-\frac{1}{x^2}) = -\cos(\frac{1}{x})$, so $f'(x) = 2x\sin(\frac{1}{x}) - \cos(\frac{1}{x})$.

Common pitfalls

- substituting $x = 0$ into the derivative formula for $x \neq 0$
- declaring no limit merely because $\sin(\frac{1}{h})$ oscillates
- omitting $-\frac{1}{x^2}$ in the chain rule

Verification

The absolute value of the origin quotient is bounded by $|h|$, which rigorously tends to 0; the two nonzero-point terms correspond exactly to the two product-rule contributions.

Method takeaway

Use the definition at exceptional points; an oscillatory factor without a limit may still yield a derivative when multiplied by a sufficiently small factor.

Extension

Examining whether $f'(x)$ tends to $f'(0)$ as $x \to 0$ distinguishes differentiability from continuity of the derivative.

Source lineage: Original synthesis / diagnosis · differentiability_and_one_sided_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Methods Classic-method variant

Q057 · Slope of an Even Function at the Origin

Problem

Suppose f is even and differentiable at $x = 0$. Prove that $f'(0) = 0$.

Answer

Using $f(-h) = f(h)$ to compare the one-sided quotients gives $f'(0) = -f'(0)$, hence $f'(0) = 0$.

Knowledge points

- symmetry of an even function
- one-sided difference quotients
- equality of one-sided limits when the derivative exists

Reading and method choice

Evenness makes the left and right function increments equal, while their denominators have opposite signs. Thus the one-sided derivatives are opposites; differentiability also requires them to be equal.

Detailed derivation

1. The right derivative is $f'_+(0) = \lim_{h\to 0^+} \frac{f(h)-f(0)}{h}$.
2. In the left derivative set $h = -t$ with $t \to 0^+ : f'_-(0) = \lim_{t\to 0^+} \frac{f(-t)-f(0)}{-t}$.
3. Evenness gives $f(-t) = f(t)$, so $f'_-(0) = -\lim_{t\to 0^+} \frac{f(t)-f(0)}{t} = -f'_+(0)$.
4. Since f is differentiable at 0, $f'_-(0) = f'_+(0) = f'(0)$. Therefore $f'(0) = -f'(0)$, and $f'(0) = 0$.

Common pitfalls

- claiming existence of the derivative from evenness alone
- losing the minus sign after substituting $h = -t$
- not using equality of the one-sided derivatives

Verification

The even functions $f(x) = x^2$ and $\cos x$ both have derivative 0 at the origin, while $|x|$ shows why the differentiability assumption cannot be removed.

Method takeaway

Symmetry controls one-sided slopes: an even function has opposite one-sided slopes at 0, so differentiability forces both to be 0.

Extension

For an odd function differentiable at 0, the derivative need not be $0; f(x) = x$ is an example.

Source lineage: Classic-method variant · derivative_definition · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.1. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Single choice | Methods | Original synthesis / diagnosis

Q058 · Recognizing the Limit of an Asymmetric Secant

Problem

Suppose f is differentiable at x_0. Which limit must equal $f'(x_0)$?

A. $\lim_{h\to 0} \frac{f(x_0+2h)-f(x_0-h)}{3h}$

B. $\lim_{h\to 0} \frac{f(x_0+2h)-f(x_0-h)}{2h}$

C. $\lim_{h\to 0} \frac{f(x_0+h)-f(x_0-h)}{h}$

D. $\lim_{h\to 0} \frac{f(x_0+2h)-f(x_0)}{h}$

Answer

A

Knowledge points

- equivalent forms of derivative quotients
- adding and subtracting a common base value
- linearity of limits

Reading and method choice

Insert and subtract $f(x_0)$ between the two asymmetric function values. This splits the expression into standard derivative quotients whose coefficients must be tracked.

Detailed derivation

1. For A, insert $f(x_0)$: $f(x_0+2h)-f(x_0-h)=[f(x_0+2h)-f(x_0)]+[f(x_0)-f(x_0-h)]$.
2. After division by 3h, the first term is $\frac{(\frac{2}{3})\cdot[f(x_0+2h)-f(x_0)]}{2h}$, whose limit is $(\frac{2}{3})f'(x_0)$.
3. The second term is $\frac{(\frac{1}{3})\cdot[f(x_0)-f(x_0-h)]}{h}=\frac{(\frac{1}{3})\cdot[f(x_0-h)-f(x_0)]}{-h}$, whose limit is $(\frac{1}{3})f'(x_0)$.
4. Their sum is $f'(x_0)$, so A is correct. The corresponding coefficients for B, C, and D are $\frac{3}{2}$, 2, and 2, respectively.

Common pitfalls

- ignoring denominator coefficients merely because $h \to 0$
- losing the two minus signs in the second term

- treating 2h as though the increment were h

Verification

For $f(x) = x$, the four limits are $1, \frac{3}{2}, 2, 2$, while $f'(x_0) = 1$; only A works.

Method takeaway

For a modified quotient, split at $f(x_0)$ and match every denominator to its actual input increment.

Extension

For $a, b > 0$, $\frac{f(x_0+ah)-f(x_0-bh)}{(a+b)h}$ also tends to $f'(x_0)$.

Source lineage: Original synthesis / diagnosis · derivative_definition · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.1. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Methods Original synthesis / diagnosis

Q059 · A Moving Corner and Parameter Classification

Problem

Let $f_{a,b}(x) = |x - a| + bx$, where $a, b \in \mathbb{R}$. Determine all pairs (a,b) for which $f_{a,b}$ is differentiable at $x = 0$, and find $f'_{a,b}(0)$.

Answer

Differentiable exactly when $a \neq 0$; then $f'_{a,b}(0) = b - \mathrm{sgn}(a)$, where $\mathrm{sgn}(a) = 1$ for $a > 0$ and $\mathrm{sgn}(a) = -1$ for $a < 0$

Knowledge points

- location of an absolute-value corner
- local removal of absolute value
- case analysis in parameters

Reading and method choice

The decisive issue is not b but whether the corner $x = a$ coincides with the test point 0. If $a \neq 0, a$ small neighborhood of 0 stays on one side of the corner. If $a = 0, a$ linear term cannot remove the slope jump of 2.

Detailed derivation

1. If $a > 0$, choose a sufficiently small neighborhood of 0 where $x - a < 0$. Then $|x - a| = a - x$, so $f_{a,b}(x) = a + (b - 1)x$ and $f'_{a,b}(0) = b - 1$.
2. If $a < 0$, choose a sufficiently small neighborhood where $x - a > 0$. Then $|x - a| = x - a$, so $f_{a,b}(x) = -a + (b + 1)x$ and $f'_{a,b}(0) = b + 1$.
3. Both cases can be written as $f'_{a,b}(0) = b - \mathrm{sgn}(a)$, and every b is allowed.
4. If $a = 0$, the quotient is $\frac{|h|+bh}{h}$. Its left limit is $b - 1$ and its right limit is $b + 1$, which always differ by 2.
5. Therefore $a \neq 0$ is necessary and sufficient, with b unrestricted and the derivative as stated.

Common pitfalls

- thinking that a suitable b can smooth the corner of $|x|$
- using the wrong local expression for $|x - a|$ when $a < 0$
- giving only a sufficient condition without excluding $a = 0$

Verification

For $a = 2$ the derivative is $b - 1$; for $a = -2$ it is $b + 1$. When $a = 0$, the one-sided derivatives differ by 2 for every b.

Method takeaway

An absolute value affects differentiability at a point only when its corner lies there; adding a linear term shifts both one-sided slopes equally.

Extension

If the test point is changed to $x = c$, differentiability holds exactly when $a \neq c$.

Source lineage: Original synthesis / diagnosis · differentiability_and_one_sided_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Methods Classic-method variant

Q060 · Derivative of an Inverse at a Corresponding Point

Problem

Let $f(x) = x^3 + x$ and $g = f^{-1}$. Given that g is differentiable at $y = 2$, find $g'(2)$.

Answer

$g'(2) = \frac{1}{4}$

Knowledge points

- corresponding points of inverse functions
- inverse-function derivative formula
- polynomial differentiation

Reading and method choice

First find x_0 satisfying $f(x_0) = 2$, then evaluate f' at that point and take the reciprocal.

Detailed derivation

1. Solve $f(x_0) = 2$, or $x_0^3 + x_0 = 2$. Since $1 + 1 = 2$, $x_0 = 1$ and $g(2) = 1$.
2. The derivative of the original function is $f'(x) = 3x^2 + 1$.
3. At the corresponding point $x_0 = 1$, $f'(1) = 3 + 1 = 4 \neq 0$.
4. The inverse-function formula gives $g'(2) = \frac{1}{f'(g(2))} = \frac{1}{f'(1)} = \frac{1}{4}$.

Common pitfalls

- substituting $y = 2$ directly into $f'(x)$
- confusing the inverse function f^{-1} with the reciprocal $\frac{1}{f}$
- failing to find the corresponding x_0 first

Verification

Locally $\Delta y \approx 4\Delta x$, so in the reverse direction $\Delta x \approx (\frac{1}{4})\Delta y$, consistent with $g'(2) = \frac{1}{4}$.

Method takeaway

An inverse derivative is a reciprocal at corresponding points: find $x_0 = g(y_0)$, then compute $\frac{1}{f'(x_0)}$.

Extension

For $g'(0)$, the corresponding point is $x_0 = 0$, giving $g'(0) = 1$.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.7. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced **Proof** **Methods** **Classic-method variant**

Q061 · Prove the Product Rule from First Principles

Problem

Suppose u and v are differentiable at x_0. Using only the definition of the derivative, prove $(uv)'(x_0) = u'(x_0)v(x_0) + u(x_0)v'(x_0)$.

Answer

$(uv)'(x_0) = u'(x_0)v(x_0) + u(x_0)v'(x_0)$

Knowledge points

- decomposition of a product increment
- differentiability implies continuity
- sum and product laws for limits

Reading and method choice

Add and subtract $u(x_0 + h)v(x_0)$ in the product increment. This produces one difference quotient for u and one for v. Differentiability also gives $u(x_0 + h) \to u(x_0)$.

Detailed derivation

1. By definition, $(uv)'(x_0) = \lim_{h \to 0} \frac{u(x_0+h)v(x_0+h) - u(x_0)v(x_0)}{h}$.
2. Add and subtract $u(x_0 + h)v(x_0)$ in the numerator to obtain $u(x_0 + h)[v(x_0 + h) - v(x_0)] + v(x_0)[u(x_0 + h) - u(x_0)]$.
3. After division by h, the quotient is $\frac{u(x_0+h) \cdot [v(x_0+h) - v(x_0)]}{h} + \frac{v(x_0) \cdot [u(x_0+h) - u(x_0)]}{h}$.
4. Since u is differentiable at x_0, it is continuous there, so $u(x_0 + h) \to u(x_0)$. The two difference quotients tend to $v'(x_0)$ and $u'(x_0)$.
5. Taking limits term by term gives $(uv)'(x_0) = u(x_0)v'(x_0) + v(x_0)u'(x_0)$, which is the desired formula.

Common pitfalls

- splitting a difference of products into a product of differences
- mismatching factors after adding the intermediate term
- using $u(x_0 + h) \to u(x_0)$ without invoking differentiability-implies-continuity

Verification

For $u(x) = x^2$ and $v(x) = x^3$, the formula gives $2x \cdot x^3 + x^2 \cdot 3x^2 = 5x^4$, agreeing with $(x^5)' = 5x^4$.

Method takeaway

A standard way to prove an operation rule is to insert a suitable intermediate term and split a complicated increment into known difference quotients.

Extension

A related decomposition, together with control of the changing denominator, proves the quotient rule from first principles.

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Advanced | Calculation | Methods | Classic-method variant

Q062 · Combined Multilayer Chain and Quotient Rules

Problem

Differentiate $y = \frac{e^{\sin(x^2)}}{1+x^3}$, and state its domain.

Answer

$y' = \frac{e^{\sin(x^2)}[2x\cos(x^2)(1+x^3)-3x^2]}{(1+x^3)^2}, x \neq -1$

Knowledge points

- composite exponential function
- multilayer chain rule
- quotient rule

Reading and method choice

The numerator $e^{\sin(x^2)}$ has two nested layers, and the denominator is a polynomial. Differentiate them separately, then apply the quotient rule.

Detailed derivation

1. Let $u(x) = e^{\sin(x^2)}$ and $v(x) = 1 + x^3$, so $y = \frac{u}{v}$ and the domain requires $x \neq -1$.
2. Differentiate u layer by layer: $u' = e^{\sin(x^2)} \cdot [\sin(x^2)]' = e^{\sin(x^2)} \cdot 2x\cos(x^2)$.
3. The denominator derivative is $v' = 3x^2$.
4. The quotient rule gives $y' = \frac{u'v-uv'}{v^2}$.
5. Substitute and factor to obtain $y' = \frac{e^{\sin(x^2)}[2x\cos(x^2)(1+x^3)-3x^2]}{(1+x^3)^2}$ for $x \neq -1$.

Common pitfalls

- omitting $\cos(x^2)$ or 2x when differentiating $e^{\sin(x^2)}$
- reversing the quotient-rule numerator
- forgetting to exclude $x = -1$

Verification

At $x = 0$ the formula gives $y'(0) = 0$. The inner function x^2 and the denominator both have derivative 0 there, which agrees with the result.

Method takeaway

Map the layers first: finish all inner chain rules before applying an outer product or quotient rule.

Extension

Writing the function as $e^{\sin(x^2)}(1 + x^3)^{-1}$ and using the product and chain rules gives the same derivative.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.5, openstax-calculus-v1-3.9. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced **Parameter / synthesis / application** **Synthesis** **Classic-method variant**

Q063 · From Average Velocity to Instantaneous Direction

Problem

A particle has position $s(t) = t^3 - 3t^2 + 2t$ meters, with t in seconds. (1) Find the average velocity on $[1, 1+h]$ for $h \neq 0$. (2) Use the definition to find $v(1)$. (3) Write the tangent to the position-time curve at $(1, s(1))$. (4) Interpret the sign of $v(1)$.

Answer

(1) $(-1+h^2)\,\mathrm{m\,s^{-1}}$; **(2)** $v(1) = -1\,\mathrm{m\,s^{-1}}$; **(3)** $s = -(t-1)$; **(4) the particle is moving in the negative direction at that instant**

Knowledge points

- average-velocity quotient
- instantaneous-velocity limit
- tangent to a position-time graph
- sign of velocity

Reading and method choice

All four parts use the same quotient. First expand $s(1+h) - s(1)$ exactly, then interpret the quotient as an average velocity, a limit, and a tangent slope.

Detailed derivation

1. Compute $s(1) = 1 - 3 + 2 = 0$ and expand $s(1+h) = (1+h)^3 - 3(1+h)^2 + 2(1+h) = -h + h^3$.
2. The average velocity on $[1, 1+h]$ is $\frac{s(1+h)-s(1)}{h} = \frac{-h+h^3}{h} = (-1+h^2)\,\mathrm{m\,s^{-1}}$.
3. Let $h \to 0$ to obtain the instantaneous velocity $v(1) = \lim_{h\to 0}(-1+h^2) = -1\,\mathrm{m\,s^{-1}}$.
4. The point on the position-time graph is $(1, 0)$ and the slope is -1, so its tangent is $s - 0 = -1(t-1)$, or $s = -(t-1)$.
5. The negative velocity means that at $t = 1$ the position is changing opposite the chosen positive coordinate direction; it does not mean that distance is negative.

Common pitfalls

- allowing only positive h and missing the two-sided instantaneous velocity
- confusing position s with speed $|v|$
- using x for the horizontal variable without explaining it

Verification

The average velocity $-1+h^2$ tends to -1 from both $h \to 0^+$ and $h \to 0^-$, so the two-sided instantaneous velocity agrees; the tangent slope is also -1.

Method takeaway

One difference quotient connects an algebraic limit, a physical rate, and a graphical tangent slope.

Extension

For $h = 0.1$ and $h = -0.1$, the average velocity is $-0.99\,\mathrm{m\,s^{-1}}$ in both cases, illustrating approximation to the instantaneous value.

Source lineage: Classic-method variant · derivative_interpretations · reference IDs: mit-18.01sc-session-3-rate, openstax-calculus-v1-3.4. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Error diagnosis | Synthesis | Original synthesis / diagnosis

Q064 · The Error in an "Infinite Derivative"

Problem

The function $f(x)=\sqrt{x}$ has domain $[0,+\infty)$. A student writes: "$f'(0)=\lim_{h\to 0}\frac{\sqrt{h}-0}{h}=\lim_{h\to 0}\frac{1}{\sqrt{h}}=+\infty$, so $f'(0)=+\infty$." Identify every key error and give the correct conclusion.

Answer

Only $h\to 0^+$ is allowed; the right-hand quotient is unbounded rather than convergent to a finite real number. Thus $f'(0)$ does not exist in the usual sense, and one must not write $f'(0)=+\infty$.

Knowledge points

- one-sided increments at a domain endpoint
- a derivative must be a finite real number
- vertical tangent versus finite derivative

Reading and method choice

There are two errors: the domain restricts the direction of the increment, and even the right-hand quotient tends to $+\infty$ rather than to a finite derivative value.

Detailed derivation

1. Because f is defined only for $x\ge 0$, an increment from the base point 0 must satisfy $h\ge 0$; only $h\to 0^+$ is admissible, not an unrestricted $h\to 0$.
2. For $h>0$, the quotient is $\frac{\sqrt{h}-0}{h}=\frac{1}{\sqrt{h}}$, which grows without bound as $h\to 0^+$.
3. The usual derivative definition requires convergence of the quotient to a finite real number; $+\infty$ records unbounded behavior and is not an acceptable derivative value.
4. Therefore f has no finite right derivative at the endpoint 0, and $f'(0)$ does not exist in the usual sense. The graph may be described as having a vertical-tangent tendency at the origin, but this does not justify $f'(0)=+\infty$.

Common pitfalls

- using negative increments that leave the domain
- treating the infinity symbol as an ordinary real number
- confusing a vertical tangent with existence of a finite derivative

Verification

For $h = 10^{-2}, 10^{-4}, 10^{-6}$, the quotients are 10,100,1000, showing unbounded growth rather than convergence to a finite number.

Method takeaway

When testing a derivative, first check the permitted direction of approach and then require a finite quotient limit.

Extension

For $f(x) = x^{\frac{1}{3}}$, the two-sided quotient at 0 is also unbounded and positive, so there is again no finite derivative.

Source lineage: Original synthesis / diagnosis · derivative_definition · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.1. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced **Multiple choice** **Synthesis** **Original synthesis / diagnosis**

Q065 · Checking Both Formula and Domain

Problem

Which differentiation formulas are correct on the indicated domains?

A. $(\ln|x|)' = \frac{1}{x}, x \neq 0$

B. $(\arccos x)' = \frac{1}{\sqrt{1-x^2}}, -1 < x < 1$

C. $(a^x)' = a^x \ln a, a > 0$ and $a \neq 1$

D. $[\sqrt{1-x^2}]' = -\frac{x}{\sqrt{1-x^2}}, -1 < x < 1$

Answer

A, C, and D.

Knowledge points

- Derivative of $\ln|x|$
- Derivative of arccosine
- Domains of exponentials and radicals

Reading and method choice

Both the formulas and their endpoint or parameter conditions must be checked. The only error is the missing negative sign in B.

Detailed derivation

1. A is correct: differentiating separately on $x > 0$ and $x < 0$ gives $\frac{1}{x}$ on both intervals.
2. B is false: the correct formula is $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$ for $-1 < x < 1$.
3. C is correct: the derivative of a positive-base exponential is itself times $\ln a$ under the stated conditions.
4. D is correct because the chain rule gives $(\frac{1}{2})(1-x^2)^{-\frac{1}{2}}(-2x) = -\frac{x}{\sqrt{1-x^2}}$.
5. Thus the correct selections are A, C, and D.

Common pitfalls

- Missing the negative sign for arccosine
- Including radical endpoints where the displayed derivative is not finite

- Forgetting the base conditions for an exponential

Verification

Because arccos x decreases as x increases, its derivative must be negative; this immediately rules out B.

Method takeaway

A basic derivative formula should be remembered together with its sign, parameter restrictions, and derivative domain.

Extension

Compare the derivatives of arcsin x and arccos x to verify that the derivative of their sum is zero.

Source lineage: Original synthesis / diagnosis · elementary_differentiation_rules · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.3. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Classic-method variant

Q066 · Chain Rule for a Four-Layer Composite

Problem

Differentiate $y = \arctan\left(e^{\sin(x^2)}\right)$, using layer variables to explain the source of every factor.

Answer

$y' = \frac{2xe^{\sin(x^2)}\cos(x^2)}{1+e^{2\sin(x^2)}}$.

Knowledge points

- Multilevel chain rule
- Derivative of arctangent
- Exponential and trigonometric composites

Reading and method choice

From outside to inside, the layers are arctangent, exponential, sine, and x^2. Naming each layer makes the four derivative factors explicit.

Detailed derivation

1. Let $u = e^{\sin(x^2)}$, so $y = \arctan u$ and $\frac{dy}{du} = \frac{1}{1+u^2}$.
2. Let $v = \sin(x^2)$, so $u = e^v$ and $\frac{du}{dv} = e^v$.
3. Let $w = x^2$, so $v = \sin w$ and $\frac{dv}{dw} = \cos w$.
4. For the innermost layer, $\frac{dw}{dx} = 2x$.
5. Multiplying along the chain gives $y' = [\frac{1}{1+u^2}]e^v \cos w \cdot 2x$; substitution yields the answer.

Common pitfalls

- Writing $1 + u$ instead of $1 + u^2$ for arctangent
- Dropping e^v after differentiating the exponential
- Missing $\cos(x^2)$ or 2x

Verification

At $x = 0$ the formula contains the factor 2x and gives $y'(0) = 0$. The original function depends on x through x^2 and is even, consistent with this value.

Method takeaway

For a deeply nested function, multiply one derivative from every layer along its dependency chain.

Extension

If x^2 is replaced by $\varphi(x)$, replace only the final factor 2x by $\varphi'(x)$.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.5, openstax-calculus-v1-3.9. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Classic-method variant

Q067 · Proving the Derivative Rule for Real Powers

Problem

Let α be any real number. Using $x^\alpha = e^{\alpha \ln x}$, prove that $(x^\alpha)' = \alpha x^{\alpha-1}$ for $x > 0$.

Answer

Write x^α as $e^{\alpha \ln x}$ and apply the chain rule: $(x^\alpha)' = \frac{e^{\alpha \ln x} \cdot \alpha}{x} = \alpha x^{\alpha-1}$.

Knowledge points

- Definition of a real power
- Exponential-logarithmic composite
- Chain rule

Reading and method choice

For an arbitrary real exponent, repeated multiplication is not an adequate definition. On $x > 0$, use the exponential-logarithmic representation and differentiate it as a composite function.

Detailed derivation

1. For $x > 0$, define $x^\alpha = e^{\alpha \ln x}$; all functions on the right are defined and differentiable.
2. Let $u(x) = \alpha \ln x$, so $x^\alpha = e^u$.
3. The chain rule gives $(e^u)' = e^u u'$, and $u' = \frac{\alpha}{x}$.
4. Hence $(x^\alpha)' = \frac{e^{\alpha \ln x} \cdot \alpha}{x} = \frac{x^\alpha \cdot \alpha}{x}$.
5. Since $\frac{x^\alpha}{x} = x^{\alpha-1}$, we obtain $(x^\alpha)' = \alpha x^{\alpha-1}$.

Common pitfalls

- Forgetting that the proof is restricted to $x > 0$
- Writing $(\alpha \ln x)'$ as $\alpha \ln x$
- Combining exponents incorrectly in the final step

Verification

For $\alpha = \frac{1}{2}$ the result is $(\sqrt{x})' = \frac{1}{2\sqrt{x}}$; for $\alpha = -1$ it is $(\frac{1}{x})' = -\frac{1}{x^2}$, matching the basic formulas.

Method takeaway

The exponential-logarithmic representation reduces differentiation of every real power to the chain rule.

Extension

For a positive differentiable $u(x)$, the same argument gives $([u(x)]^{\alpha})' = \alpha u^{\alpha-1}u'$.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.9. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Parameter / synthesis / application | Synthesis | Classic-method variant

Q068 · Differentiability Classification for a Three-Parameter Piecewise Function

Problem

Let $f(x) = \begin{cases} x^2 + ax + b, & x < 1, \\ c \ln x + 2, & x \geq 1. \end{cases}$ Find all real triples (a, b, c) for which f is differentiable at $x = 1$, and give $f'(1)$.

Answer

$(a, b, c) = (t, 1 - t, t + 2)$, where $t \in \mathbb{R}$; and $f'(1) = t + 2$.

Knowledge points

- Continuity at a joining point
- Equality of one-sided derivatives
- Parameterization of a solution set

Reading and method choice

The three parameters are constrained by only two independent equations—continuity and equal one-sided derivatives—so the answer is a family rather than one triple.

Detailed derivation

1. The right-hand piece gives $f(1) = c \ln 1 + 2 = 2$.
2. The left-hand limit is $1 + a + b$, so continuity requires $1 + a + b = 2$, or $a + b = 1$.
3. The left derivative is $2x + a$, which equals $2 + a$ at $x = 1$.
4. The right derivative is $\frac{c}{x}$, which equals c at $x = 1$; differentiability requires $c = 2 + a$.
5. Set $a = t$. Then $b = 1 - t$ and $c = t + 2$, where t is any real number.
6. For these parameters, the common one-sided derivative is $t + 2$, so $f'(1) = t + 2$.

Common pitfalls

- Treating $\ln 1$ as 1
- Assuming three parameters must have a unique solution
- Checking continuity without checking one-sided derivatives

Verification

After substitution, the left value is $1+t+1-t=2$ and the right value is 2; the left derivative $2+t$ equals the right derivative $c=t+2$.

Method takeaway

When there are more parameters than independent conditions, state the free parameter and the complete solution family.

Extension

If $f'(1)=0$ is also required, then $t=-2$ and the unique triple is $(a,b,c)=(-2,3,0)$.

Source lineage: Classic-method variant · elementary_differentiation_rules · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.3. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Classic-method variant

Q069 · Deriving the Derivative of Arctangent from the Inverse Formula

Problem

View arctan x as the inverse of tan y on $(-\frac{\pi}{2}, \frac{\pi}{2})$, and prove that $(\arctan x)' = \frac{1}{1+x^2}$.

Answer

If $y = \arctan x$**, the inverse formula gives** $y' = \frac{1}{\sec^2 y} = \cos^2 y = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}$**.**

Knowledge points

- Inverse-function derivative formula
- Tangent and arctangent
- Identity $1 + \tan^2 y = \sec^2 y$

Reading and method choice

Tangent is one-to-one on the specified interval and its derivative $\sec^2 y$ is never zero there. Apply the inverse derivative formula and then express the result entirely in x.

Detailed derivation

1. Let $F(y) = \tan y$ with domain restricted to $(-\frac{\pi}{2}, \frac{\pi}{2})$; its inverse is $g(x) = \arctan x$.
2. On this interval, $F'(y) = \sec^2 y > 0$, so the inverse derivative formula applies.
3. If $y = g(x) = \arctan x$, then $g'(x) = \frac{1}{F'(y)} = \frac{1}{\sec^2 y}$.
4. Use $\frac{1}{\sec^2 y} = \cos^2 y = \frac{1}{1+\tan^2 y}$.
5. Since $\tan y = x$, we obtain $g'(x) = \frac{1}{1+x^2}$ for every real x.

Common pitfalls

- Failing to restrict tangent to a one-to-one interval
- Writing the inverse derivative as $\sec^2 x$
- Failing to replace y by an expression in x

Verification

At $x = 0$, the arctangent graph has slope 1, and the formula gives $\frac{1}{1+0} = 1$.

Method takeaway

An inverse-trigonometric derivative is obtained by taking the reciprocal of the original trigonometric derivative at the corresponding point.

Extension

Use the same method with sine on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ to derive the derivative of arcsine.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.7. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Classic-method variant

Q070 · The nth Derivative of the Natural Logarithm

Problem

For $x > 0$, compute the first four derivatives of ln x and then state its nth derivative for $n \geq 1$.

Answer

$(\ln x)^{(n)} = \frac{(-1)^{n-1}(n-1)!}{x^n}$.

Knowledge points

- Successive derivatives of logarithm
- Factorial pattern
- Alternating signs

Reading and method choice

With each differentiation, the denominator power increases by one, the sign alternates, and the coefficients form $0!$, $1!$, $2!$, and $3!$.

Detailed derivation

1. The first derivative is $(\ln x)' = \frac{1}{x}$.
2. The second and third derivatives are $-\frac{1}{x^2}$ and $\frac{2}{x^3}$.
3. The fourth derivative is $-\frac{6}{x^4}$.
4. The nth sign is $(-1)^{n-1}$, the denominator is x^n, and the coefficient is $(n-1)!$.
5. Thus $(\ln x)^{(n)} = \frac{(-1)^{n-1}(n-1)!}{x^n}$ for $n \geq 1$.

Common pitfalls

- Using $n!$ instead of $(n-1)!$
- Using $(-1)^n$ for the sign
- Shifting the denominator power away from the derivative order

Verification

At $n = 1$ the formula gives $\frac{1}{x}$, and at $n = 4$ it gives $-\frac{3!}{x^4} = -\frac{6}{x^4}$, matching the direct calculations.

Method takeaway

A few low-order derivatives reveal the sign, factorial, and power patterns simultaneously.

Extension

For $\ln(ax + b)$, each order also contributes a^n and replaces x by $ax + b$.

Source lineage: Classic-method variant · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Classic-method variant

Q071 · A Unified nth-Derivative Formula for an Exponential-Cosine Product

Problem

Let $a^2 + b^2 > 0, r = \sqrt{a^2 + b^2}$, and choose θ so that $\cos\theta = \frac{a}{r}$ and $\sin\theta = \frac{b}{r}$. Prove by induction that $y = e^{ax}\cos(bx + \varphi)$ satisfies $y^{(n)} = r^n e^{ax}\cos(bx + \varphi + n\theta)$.

Answer

The formula holds for every integer $n \geq 0$; the induction step uses $a\cos\psi - b\sin\psi = r\cos(\psi + \theta)$.

Knowledge points

- Exponential-trigonometric products
- Auxiliary-angle identity
- Mathematical induction

Reading and method choice

One differentiation increases the phase by the fixed angle θ and multiplies the amplitude by the fixed factor r. Establishing this update proves the general formula inductively.

Detailed derivation

1. For $n = 0$, the right side is $e^{ax}\cos(bx + \varphi) = y$, so the base case holds.
2. From $\cos\theta = \frac{a}{r}$ and $\sin\theta = \frac{b}{r}$, $a\cos\psi - b\sin\psi = r\cos(\psi + \theta)$.
3. Assume $y^{(k)} = r^k e^{ax}\cos(bx + \varphi + k\theta)$.
4. Set $\psi = bx + \varphi + k\theta$ and differentiate the induction hypothesis to get $y^{(k+1)} = r^k e^{ax}[a\cos\psi - b\sin\psi]$.
5. The auxiliary-angle identity gives $y^{(k+1)} = r^{k+1}e^{ax}\cos(bx + \varphi + (k+1)\theta)$.
6. Thus the induction step holds, proving the formula for all $n \geq 0$.

Common pitfalls

- Choosing the wrong sign for $\sin\theta$
- Missing the factor r at each differentiation
- Failing to exclude $a = b = 0$, for which θ is not defined as stated

Verification

At $n = 1$, the formula gives $re^{ax}\cos(bx + \varphi + \theta) = e^{ax}[a\cos(bx + \varphi) - b\sin(bx + \varphi)]$, exactly the direct derivative.

Method takeaway

For some closed function families, successive differentiation produces a fixed amplitude scaling and phase shift.

Extension

Replacing cosine by sine gives $y^{(n)} = r^n e^{ax}\sin(bx + \varphi + n\theta)$.

Source lineage: Classic-method variant · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Parameter / synthesis / application | Synthesis | Classic-method variant

Q072 · The General Derivative of $x^2 \sin x$

Problem

Let $y = x^2 \sin x$. For $n \geq 2$, find an explicit formula for $y^{(n)}$ and check it at $n = 2$.

Answer

$$y^{(n)} = x^2 \sin\Big(x + \frac{n\pi}{2}\Big) + 2nx \sin\Big(x + \frac{(n-1)\pi}{2}\Big) + n(n-1) \sin\Big(x + \frac{(n-2)\pi}{2}\Big).$$

Knowledge points

- Higher derivative of a product
- Termination of polynomial derivatives
- Sine derivative cycle

Reading and method choice

Distribute the n differentiations between x^2 and $\sin x$. Because the third and higher derivatives of x^2 vanish, the full sum reduces to only three terms.

Detailed derivation

1. The higher product formula gives $y^{(n)} = \sum_{k=0}^{n} C(n,k)(x^2)^{(k)}(\sin x)^{(n-k)}$.
2. Since $(x^2)^{(0)} = x^2, (x^2)' = 2x, (x^2)'' = 2$, and derivatives for $k \geq 3$ vanish, retain only $k = 0, 1, 2$.
3. Use $(\sin x)^{(m)} = \sin(x + \frac{m\pi}{2})$.
4. The three terms are $x^2 \sin\Big(x + \frac{n\pi}{2}\Big)$, $2nx \sin\Big(x + \frac{(n-1)\pi}{2}\Big)$, and $\binom{n}{2} 2 \sin\Big(x + \frac{(n-2)\pi}{2}\Big)$.
5. Because $2C(n,2) = n(n-1)$, these terms give the stated formula.
6. At $n = 2$ it becomes $-x^2 \sin x + 4x \cos x + 2 \sin x$, matching direct differentiation.

Common pitfalls

- Switching k and $n - k$ without adjusting the phase
- Omitting the binomial coefficients
- Simplifying $2C(n, 2)$ incorrectly

Verification

Directly, $y' = 2x \sin x + x^2 \cos x$ and $y'' = 2 \sin x + 4x \cos x - x^2 \sin x$, exactly the $n = 2$ specialization.

Method takeaway

For a polynomial times another function, the higher-product sum terminates because sufficiently high polynomial derivatives vanish.

Extension

Replacing x^2 by x^m leaves $k = 0, 1, \ldots, m$, for a total of $m + 1$ terms.

Source lineage: Classic-method variant · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Classic-method variant

Q073 · Decompose Before Taking Higher Derivatives of a Rational Function

Problem

Let $y = \frac{1}{(x-1)(x+2)}, x \neq 1, -2$. Find $y^{(n)}$ for $n \geq 0$.

Answer

$y^{(n)} = \frac{(-1)^n n!}{3} \cdot [\frac{1}{(x-1)^{n+1}} - \frac{1}{(x+2)^{n+1}}]$.

Knowledge points

- Partial-fraction decomposition
- Higher derivatives of reciprocal linear factors
- Linearity

Reading and method choice

Repeated quotient rules would become complicated. First split the rational function into two reciprocal linear factors, then apply one standard formula termwise.

Detailed derivation

1. Write $\frac{1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$.
2. Setting $x = 1$ gives $A = \frac{1}{3}$, and setting $x = -2$ gives $B = -\frac{1}{3}$.
3. Thus $y = (\frac{1}{3})[\frac{1}{x-1} - \frac{1}{x+2}]$.
4. Use $[\frac{1}{x-a}]^{(n)} = \frac{(-1)^n n!}{(x-a)^{n+1}}$.
5. Differentiate termwise to obtain $y^{(n)} = \frac{(-1)^n n!}{3} \cdot [\frac{1}{(x-1)^{n+1}} - \frac{1}{(x+2)^{n+1}}]$.

Common pitfalls

- Reversing a partial-fraction sign
- Confusing the shifts in the two denominators
- Omitting the $+1$ in the denominator power

Verification

At $n = 0$, the formula returns the partial-fraction expression, which combines back to the original function.

Method takeaway

For a rational function with linear factors, algebraic decomposition before differentiation is often the key to an nth-order formula.

Extension

Repeated linear factors require terms such as $\frac{1}{(x-a)^2}$, each handled by the negative-integer power rule.

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Advanced Calculation Synthesis Classic-method variant

Q074 · Second Derivative of a Quadratic Implicit Curve

Problem

The curve $x^2 + xy + y^2 = 7$ passes through $(1, 2)$. Find y' and y'' at that point.

Answer

$y' = -\frac{4}{5}$ **and** $y'' = -\frac{42}{125}$.

Knowledge points

- First implicit derivative
- Differentiating the derivative relation again
- Product rule
- Point substitution

Reading and method choice

Keep the first derivative relation and differentiate the entire relation again. Both xy' and $2yy'$ require the product rule.

Detailed derivation

1. The first derivative relation is $2x + y + xy' + 2yy' = 0$, so $y' = -\frac{2x+y}{x+2y}$.
2. At $(1, 2)$, $y' = -\frac{2+2}{1+4} = -\frac{4}{5}$.
3. Differentiate the relation again to get $2 + y' + y' + xy'' + 2(y')^2 + 2yy'' = 0$.
4. Combine terms: $2 + 2y' + 2(y')^2 + (x + 2y)y'' = 0$.
5. Substitute $x = 1, y = 2, y' = -\frac{4}{5}$ to obtain $2 - \frac{8}{5} + \frac{32}{25} + 5y'' = 0$.
6. The first three terms equal $\frac{42}{25}$, so $y'' = -\frac{42}{125}$.

Common pitfalls

- Omitting $(y')^2$ in the second differentiation
- Writing $\frac{d(xy')}{dx}$ as only xy''
- Discarding the first derivative relation too early

Verification

$(1,2)$ satisfies the curve, and $\frac{42}{25}+5(-\frac{42}{125})=0$ in the second derivative relation.

Method takeaway

For a second implicit derivative, differentiate the uncompressed first derivative relation and retain every product-rule term.

Extension

The general form is $y''=-\frac{2+2y'+2(y')^2}{x+2y}$ wherever the denominator is nonzero.

Source lineage: Classic-method variant · implicit_differentiation · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.8. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Classic-method variant

Q075 · Second Derivative of a Parametric Curve

Problem

For $x = t^2 + 1$ and $y = t^3$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $t = 1$.

Answer

$\frac{dy}{dx} = \frac{3}{2}$ **and** $\frac{d^2y}{dx^2} = \frac{3}{4}$**.**

Knowledge points

- First parametric derivative
- Second parametric derivative formula
- Differentiate with respect to t and divide by $\frac{dx}{dt}$

Reading and method choice

The second derivative is not $\frac{\frac{d^2y}{dt^2}}{\frac{d^2x}{dt^2}}$. First differentiate $\frac{dy}{dx}$ as a function of t, then divide by $\frac{dx}{dt}$.

Detailed derivation

1. $\frac{dx}{dt} = 2t$ and $\frac{dy}{dt} = 3t^2$.
2. For $t \neq 0$, $\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3t}{2}$.
3. Differentiate with respect to t: $\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{3}{2}$.
4. Use $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{\frac{3}{2}}{2t} = \frac{3}{4t}$.
5. At $t = 1$, $\frac{dy}{dx} = \frac{3}{2}$ and $\frac{d^2y}{dx^2} = \frac{3}{4}$.

Common pitfalls

- Using $\frac{\frac{d^2y}{dt^2}}{\frac{d^2x}{dt^2}}$
- Canceling t without noting $t \neq 0$
- Forgetting the second division by $\frac{dx}{dt}$

Verification

Eliminating t gives $y = (x-1)^{\frac{3}{2}}$; at $x = 2$, ordinary differentiation gives $\frac{3}{2}$ and $\frac{3}{4}$.

Method takeaway

For a second parametric derivative: find $\frac{dy}{dx}$, differentiate it with respect to t, then divide by $\frac{dx}{dt}$.

Extension

Equivalently, $\frac{d^2y}{dx^2} = \frac{x'(t)y''(t)-y'(t)x''(t)}{[x'(t)]^3}$ when $x'(t) \neq 0$.

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Advanced | Multiple choice | Synthesis | Original synthesis / diagnosis

Q076 · Modeling Principles for Related Rates

Problem

Which statements about related-rate problems are correct? Select all that apply.

A. First write a geometric or physical relation valid over time, then differentiate with respect to t

B. If a length is decreasing, its rate is negative under the chosen positive direction

C. The unit of $\frac{dV}{dt}$ is $\frac{\text{length}^3}{\text{time}}$

D. Current numerical values may be substituted before differentiation and treated as constants because only one instant matters

Answer

A, B, and C

Knowledge points

- Order of related-rate modeling
- Signed rates
- Dimensional checking
- Variables versus instantaneous values

Reading and method choice

Every geometric quantity is a function of time. Instantaneous values should be inserted only after differentiating, or a changing quantity may be frozen incorrectly.

Detailed derivation

1. A is correct: a relation valid over time allows the chain rule to connect the rates.
2. B is correct: a decreasing quantity has a negative signed rate under the chosen positive direction.
3. C is correct because volume has dimension length^3, so its rate has dimension $\frac{\text{length}^3}{\text{time}}$.
4. D is false: substituting first can turn a changing variable into a constant and erase its rate.
5. Therefore A, B, and C are correct.

Common pitfalls

- Ignoring the sign of a directed rate
- Substituting instantaneous values too early
- Giving an area or volume rate the unit length/time

Verification

For $V = (\frac{4}{3})\pi r^3$, substituting $r = 5$ first makes V look constant and falsely gives derivative 0; differentiating first gives $\frac{dV}{dt} = \frac{4\pi r^2 dr}{dt}$.

Method takeaway

Use the fixed sequence: relation, differentiate in time, substitute instantaneous data, then check sign and units.

Extension

Choosing the opposite positive direction changes some signs, but consistent use leaves the physical conclusion unchanged.

Source lineage: Original synthesis / diagnosis · related_rates · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.1. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced **Parameter / synthesis / application** **Synthesis** **Classic-method variant**

Q077 · Radius and Volume Rates in a Conical Water Surface

Problem

Water in an inverted conical tank forms similar cones satisfying $h = 3r$, with h, r measured in cm. The depth increases at $\frac{dh}{dt} = 6\,\text{cm}\,\text{min}^{-1}$. When $h = 6$ cm, find $\frac{dr}{dt}$ and $\frac{dV}{dt}$.

Answer

$\frac{dr}{dt} = 2\,\text{cm}\,\text{min}^{-1}$; $\frac{dV}{dt} = 24\pi\,\text{cm}^3\,\text{min}^{-1}$.

Knowledge points

- Similarity relation
- Cone volume
- Related rates with several variables
- Unit checking

Reading and method choice

Use similarity to connect r and h. Expressing volume as a function of h alone avoids two unknown rates.

Detailed derivation

1. From $h = 3r$, $r = \frac{h}{3}$. Differentiate to get $\frac{dr}{dt} = \frac{1}{3}\frac{dh}{dt} = 2\,\text{cm}\,\text{min}^{-1}$.
2. When $h = 6$ cm, $r = \frac{6}{3} = 2$ cm.
3. The water volume is $V = (\frac{1}{3})\pi r^2 h$. Substituting $r = \frac{h}{3}$ gives $V = \frac{\pi h^3}{27}$.
4. Differentiate in time: $\frac{dV}{dt} = \frac{(\frac{\pi h^2}{9})\cdot dh}{dt}$.
5. With $h = 6$ and $\frac{dh}{dt} = 6$, $\frac{dV}{dt} = \left(\frac{\pi\cdot 36}{9}\right)6 = 24\pi\,\text{cm}^3\,\text{min}^{-1}$.
6. Both rates are positive, consistent with increasing depth and volume.

Common pitfalls

- Treating the tank's rim radius as the current water radius
- Omitting $h = 3r$

- Dropping the factor $\frac{1}{3}$ in cone volume
- Using cm min^{-1} for a volume rate

Verification

Differentiating $V = (\frac{1}{3})\pi r^2 h$ directly and substituting $r = 2, h = 6, r' = 2, h' = 6$ also gives 24π.

Method takeaway

For similar-container related rates, reduce the variables with geometry before differentiating in time.

Extension

If $h = kr$, then $V = \frac{\pi h^3}{3k^2}$, and the same method gives its rate.

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Advanced | Parameter / synthesis / application | Synthesis | Open-text method adaptation

Q078 · Linear and Angular Rates of a Sliding Ladder

Problem

A 10 m ladder leans against a wall. Its foot moves away from the wall at $\frac{dx}{dt} = 0.6\,\mathrm{m\,s^{-1}}$. Let x be the foot's distance from the wall, y the top height, and θ the angle with the ground. When $x = 6$ m, find $\frac{dy}{dt}$ and $\frac{d\theta}{dt}$.

Answer

$\frac{dy}{dt} = -0.45\,\mathrm{m\,s^{-1}}$; $\frac{d\theta}{dt} = -0.075\,\mathrm{rad\,s^{-1}}$.

Knowledge points

- Pythagorean relation
- Signs in related rates
- Angular velocity
- Trigonometric relation

Reading and method choice

The fixed length gives $x^2 + y^2 = 100$. The angular rate follows from $x = 10\cos\theta$. Negative signs should match a falling top and decreasing angle.

Detailed derivation

1. When $x = 6$, $x^2 + y^2 = 100$ gives $y = \sqrt{100 - 36} = 8$ m.
2. Differentiate $x^2 + y^2 = 100$ in time: $\frac{2xdx}{dt} + \frac{2ydy}{dt} = 0$.
3. Substitute $x = 6$, $y = 8$, and $\frac{dx}{dt} = 0.6$ to get $\frac{dy}{dt} = -\frac{6}{8}(0.6) = -0.45\,\mathrm{m\,s^{-1}}$.
4. From $x = 10\cos\theta$, differentiation gives $\frac{dx}{dt} = -\frac{10\sin\theta \cdot d\theta}{dt}$.
5. Since $\sin\theta = \frac{y}{10} = 0.8$, $\frac{d\theta}{dt} = \frac{0.6}{-10(0.8)} = -0.075\,\mathrm{rad\,s^{-1}}$.
6. The two negative signs indicate that the top falls and the angle decreases.

Common pitfalls

- Treating the 10 m length as variable
- Taking a negative height

- Omitting the angular unit rad s^{-1}
- Ignoring the physical meaning of signs

Verification

Using $y = 10 \sin \theta$ gives $\frac{dy}{dt} = 10 \cos \theta \cdot \theta' = 10(0.6)(-0.075) = -0.45\,\mathrm{m\,s^{-1}}$, confirming the result.

Method takeaway

Check a related-rate answer with a second relation, its sign, and its units.

Extension

If the foot moves toward the wall, $\frac{dx}{dt} < 0$ and both the top rate and angular rate reverse sign.

Source lineage: Open-text method adaptation · related_rates · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.1. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Proof | Synthesis | Classic-method variant

Q079 · A General Derivative Formula for a Separable Implicit Equation

Problem

Let φ and ψ be differentiable, and suppose $y = y(x)$ satisfies $\varphi(x) + \psi(y) = C$ with $\psi'(y) \neq 0$. Prove $y' = -\frac{\varphi'(x)}{\psi'(y)}$, and explain the role of $\psi'(y) \neq 0$.

Answer

Differentiation gives $\varphi'(x) + \psi'(y)y' = 0$. Since $\psi'(y) \neq 0$, division yields $y' = -\frac{\varphi'(x)}{\psi'(y)}$.

Knowledge points

- Implicit composite differentiation
- Chain rule
- Nonzero condition
- Proof of a general formula

Reading and method choice

The term $\psi(y)$ is a composition of ψ with $y(x)$. The only division in the proof requires $\psi'(y)$ to be nonzero.

Detailed derivation

1. Start with the identity $\varphi(x) + \psi(y(x)) = C$.
2. Differentiate $\varphi(x)$ to obtain $\varphi'(x)$.
3. By the chain rule, $\frac{d[\psi(y(x))]}{dx} = \psi'(y)y'$, while $\frac{dC}{dx} = 0$.
4. Therefore $\varphi'(x) + \psi'(y)y' = 0$.
5. Since $\psi'(y) \neq 0$, divide to get $y' = -\frac{\varphi'(x)}{\psi'(y)}$.
6. If $\psi'(y) = 0$, this division fails; a finite y' may not exist, and vertical behavior may require separate analysis.

Common pitfalls

- Omitting y' from $\frac{d[\psi(y)]}{dx}$
- Failing to use or explain the nonzero condition
- Treating C as a variable function

Verification

For $\varphi(x) = x^2, \psi(y) = y^2$, and $C = 25$, the formula gives $y' = -\frac{2x}{2y} = -\frac{x}{y}$, matching the circle calculation.

Method takeaway

The general implicit formula comes from one chain-rule differentiation; the nonzero denominator permits a finite local slope.

Extension

If x is differentiated with respect to y and $\varphi'(x) \neq 0$, symmetry gives $\frac{dx}{dy} = -\frac{\psi'(y)}{\varphi'(x)}$.

Source lineage: Classic-method variant · implicit_differentiation · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.8. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Calculation | Synthesis | Open-text method adaptation

Q080 · Absolute and Relative Error Estimates for a Circle's Area

Problem

A circle's radius is measured as $r = 10\,\text{cm}$ with possible absolute measurement error $|dr| \le 0.02\,\text{cm}$. Use differentials to estimate the maximum absolute and relative errors in $A = \pi r^2$.

Answer

$|dA| \approx 0.4\pi\,\text{cm}^2$; $\frac{|dA|}{A} \approx 0.004 = 0.4\%$.

Knowledge points

- Absolute differential estimate
- Relative differential
- Error units
- Percentage

Reading and method choice

Use $dA = 2\pi r\,dr$. Take absolute values with the maximum $|dr|$, then divide by A for the relative error.

Detailed derivation

1. For $A = \pi r^2$, the differential is $dA = 2\pi r\,dr$.
2. At $r = 10$ cm with $|dr| \le 0.02$ cm, $|dA| \approx 2\pi \cdot 10 \cdot 0.02 = 0.4\pi$ cm^2.
3. The relative differential is $\frac{dA}{A} = \frac{2\pi r\,dr}{\pi r^2} = \frac{2\,dr}{r}$.
4. Thus the maximum relative estimate is $\frac{2|dr|}{r} = \frac{2\cdot 0.02}{10} = 0.004$.
5. As a percentage, $0.004 = 0.4\%$.

Common pitfalls

- Forgetting absolute values for maximum error
- using cm instead of cm^2
- Writing 0.004 as 4%
- Using diameter in place of radius

Verification

The dimension of $2\pi r\,dr$ is cm^2, while relative error is dimensionless. A 0.2% radius error produces approximately a 0.4% area error.

Method takeaway

For $y = x^n$, $\frac{dy}{y} = \frac{n \cdot dx}{x}$, which quickly shows the relative-error amplification factor.

Extension

For $V = (\frac{4}{3})\pi r^3$, the relative error estimate is $\frac{|dV|}{V} \approx \frac{3|dr|}{r}$.

Source lineage: Open-text method adaptation · differentials_and_linear_approximation · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Classic-method variant

Q081 · Approximating a Reciprocal Square Root

Problem

Use differential linearization at $x_0 = 1$ to approximate $\frac{1}{\sqrt{0.98}}$, and state whether this method provides a rigorous error bound.

Answer

$\frac{1}{\sqrt{0.98}} \approx 1.01$; **differential linearization in this chapter gives a first-order approximation but no automatic rigorous error bound.**

Knowledge points

- Differential of a negative power
- Linear approximation
- Approximation notation
- Method boundary

Reading and method choice

Let $f(x) = x^{-\frac{1}{2}}$. The target 0.98 differs from the base point 1 by -0.02.

Detailed derivation

1. Take $f(x) = x^{-\frac{1}{2}}, x_0 = 1$, and $\Delta x = 0.98 - 1 = -0.02$.
2. $f(1) = 1$ and $f'(x) = -(\frac{1}{2})x^{-\frac{3}{2}}$.
3. Thus $f'(1) = -\frac{1}{2}$ and $dy = f'(1)\Delta x = (-\frac{1}{2})(-0.02) = 0.01$.
4. Linearization gives $f(0.98) \approx f(1) + dy = 1.01$.
5. Therefore $\frac{1}{\sqrt{0.98}} \approx 1.01$. First-order linearization alone does not supply a rigorous error bound.

Common pitfalls

- Missing the negative sign of Δx
- Differentiating the negative power incorrectly
- Using equality instead of approximation
- Inventing an unproved remainder bound

Verification

$1.01^2 \cdot 0.98 \approx 0.999698$, close to 1, supporting the approximation; this numerical check is not a rigorous error proof.

Method takeaway

Differentials give a local linear prediction; numerical checks assess plausibility, while rigorous error control requires additional theory.

Extension

The same linearization gives $\frac{1}{\sqrt{1.03}} \approx 0.985$.

Source lineage: Classic-method variant · differentials_and_linear_approximation · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Parameter / synthesis / application | Synthesis | Classic-method variant

Q082 · Linear Prediction and Actual Discrepancy for a Fifth Power

Problem

Let $y = x^5$. Using $x_0 = 1$ and $dx = 0.02$, approximate $(1.02)^5$ by differentials and give the predicted relative increment. Then use the exact value 1.1040808032 from direct multiplication to find exact minus approximate, and explain why this is not an error bound supplied by this chapter.

Answer

$(1.02)^5 \approx 1.10$; the relative increment is about $0.10 = 10$%; exact minus approximate is 0.0040808032.

Knowledge points

- Linearization of a power
- Relative differential
- Approximation discrepancy
- Boundary of error claims

Reading and method choice

The base value is 1, so the numerical absolute and relative differentials coincide. A post hoc discrepancy at one input is not a uniform bound over a neighborhood.

Detailed derivation

1. For $y = x^5, dy = 5x^4 dx$. At $x_0 = 1$ with $dx = 0.02, dy = 5 \cdot 1^4 \cdot 0.02 = 0.10$.
2. The linear approximation is $(1.02)^5 \approx 1^5 + 0.10 = 1.10$.
3. The relative differential is $\frac{dy}{y} = \frac{5dx}{x} = 5 \cdot 0.02 = 0.10$, or about 10%.
4. Using the given exact value, exact minus approximate is $1.1040808032 - 1.10 = 0.0040808032$.
5. This difference describes only this input and does not show that all nearby errors are below the same number.
6. Linearization in this chapter supplies no uniform remainder estimate here, so 1.10 remains an approximation without a rigorous bound.

Common pitfalls

- Writing 10% as 0.10%
- Using equality for 1.10

- Treating a post hoc discrepancy as a general bound
- Forgetting that the base value is 1

Verification

Direct multiplication gives $(1.02)^2 = 1.0404, (1.02)^4 = 1.08243216$, and then 1.1040808032.

Method takeaway

A differential gives a first-order prediction; the observed discrepancy evaluates one case but does not replace an error theory.

Extension

With $dx = 0.01$, the linear prediction is 1.05; direct multiplication can be used to observe how the discrepancy changes.

Source lineage: Classic-method variant · differentials_and_linear_approximation · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | True/false with justification | Synthesis | Classic-method variant

Q083 · Complete Classification of a Parameterized Power Corner

Problem

Decide whether the following statement is true and prove it: for $\alpha > 0$, $f_\alpha(x) = |x|^\alpha$ is differentiable at $x = 0$ if and only if $\alpha > 1$; when differentiable, $f'_\alpha(0) = 0$.

Answer

True.

Knowledge points

- parameter classification
- definition of a derivative
- power limits
- one-sided quotients

Reading and method choice

The origin quotient is exactly $\frac{|h|^\alpha}{h}$. Splitting according to whether $\alpha - 1$ is positive, zero, or negative proves both sufficiency and necessity.

Detailed derivation

1. Since $f_\alpha(0) = 0$, the origin quotient is $\frac{|h|^\alpha}{h} = \operatorname{sgn}(h)|h|^{\alpha-1}$.
2. If $\alpha > 1$, its absolute value is $|h|^{\alpha-1} \to 0$, so the two-sided limit exists and $f'_\alpha(0) = 0$.
3. If $\alpha = 1$, the quotient is 1 for $h \to 0^+$ and -1 for $h \to 0^-$, so the one-sided limits differ.
4. If $0 < \alpha < 1$, the quotient $h^{\alpha-1}$ grows without bound for $h \to 0^+$ and equals $-|h|^{\alpha-1}$ for $h \to 0^-$, so no finite two-sided limit exists.
5. These cases exhaust $\alpha > 0$, proving differentiability exactly for $\alpha > 1$, with derivative 0.

Common pitfalls

- proving only sufficiency for $\alpha > 1$
- ignoring the opposite signs when $\alpha = 1$
- treating an unbounded quotient as a finite derivative

Verification

For $\alpha = 2$ the function is x^2 and has derivative 0 at 0; $\alpha = 1$ gives the corner $|x|$; $\alpha = \frac{1}{2}$ gives an even steeper radical corner, matching the classification.

Method takeaway

In a parameterized differentiability problem, a critical exponent in the difference quotient often controls the answer; here it is $\alpha = 1$.

Extension

For $c|x|^\alpha$ with $c \neq 0$, the differentiability range is unchanged; derivatives away from the corner are simply scaled by c.

Source lineage: Classic-method variant · differentiability_and_one_sided_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Parameter / synthesis / application | Synthesis | Original synthesis / diagnosis

Q084 · Stable Secants but Uncontrolled Nearby Tangents

Problem

Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x^2}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$ (1) Prove continuity at 0. (2) Prove differentiability at 0 and write the tangent there. (3) Find $f'(x)$ for $x \neq 0$. (4) Construct $x_n \to 0^+$ showing that f' is not continuous at 0.

Answer

f is continuous at 0 with $f'(0) = 0$ and tangent $y = 0$; for $x \neq 0$, $f'(x) = 2x\sin(\frac{1}{x^2}) - (\frac{2}{x})\cos(\frac{1}{x^2})$; one may take $x_n = (\frac{1}{2\pi n})^{\frac{1}{2}}$

Knowledge points

- bounded oscillatory factor
- first-principles differentiation
- chain rule
- disproving a limit by a sequence

Reading and method choice

The function value and the origin quotient are controlled by x^2 and $|x|$, so continuity and differentiability hold. However, the nonzero-point derivative contains a $\frac{1}{x}$ amplification, which becomes unbounded along a selected sequence.

Detailed derivation

1. Since $|f(x)| = |x^2 \sin(\frac{1}{x^2})| \leq x^2 \to 0 = f(0)$, f is continuous at 0.
2. The origin quotient is $\frac{f(h)}{h} = h\sin(\frac{1}{h^2})$, whose absolute value is at most $|h|$. Thus $f'(0) = 0$, and the tangent through $(0, 0)$ is $y = 0$.
3. For $x \neq 0$, the product and chain rules give $f'(x) = 2x\sin(\frac{1}{x^2}) + x^2\cos(\frac{1}{x^2})(-\frac{2}{x^3}) = 2x\sin(\frac{1}{x^2}) - (\frac{2}{x})\cos(\frac{1}{x^2})$.
4. Take $x_n = (\frac{1}{2\pi n})^{\frac{1}{2}}$. Then $x_n \to 0^+$, $\frac{1}{x_n^2} = 2\pi n$, $\sin(2\pi n) = 0$, and $\cos(2\pi n) = 1$.
5. Hence $f'(x_n) = -\frac{2}{x_n} \to -\infty$, so it cannot tend to $f'(0) = 0$. Therefore f' is not continuous at 0 even though the tangent there exists.

Common pitfalls

- declaring f discontinuous merely because $\sin(\frac{1}{x^2})$ oscillates
- substituting 0 into the derivative formula valid only for $x \neq 0$
- claiming oscillation without supplying a verifiable sequence

Verification

The secant slope $\frac{f(x)}{x} = x\sin(\frac{1}{x^2})$ is bounded by $|x|$ and tends to 0, while the nearby tangent slopes along the constructed sequence tend to $-\infty$; these facts are compatible.

Method takeaway

Differentiability at one point controls secants ending at that point, not convergence of derivatives at nearby points.

Extension

Compare $x^2\sin(\frac{1}{x})$ with the present function and observe how changing the oscillation rate changes the amplification term in the nonzero-point derivative.

Source lineage: Original synthesis / diagnosis · derivative_interpretations · reference IDs: mit-18.01sc-session-3-rate, openstax-calculus-v1-3.4. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Calculation | Synthesis | Classic-method variant

Q085 · Structural Simplification after Nested Rules

Problem

Differentiate $y = \arctan[\frac{e^x-1}{e^x+1}]$, and simplify the result so that it contains no compound fraction.

Answer

$y' = \frac{e^x}{e^{2x}+1}$

Knowledge points

- derivative of arctangent
- derivative of the exponential
- quotient rule
- algebraic identity simplification

Reading and method choice

Set $u = \frac{e^x-1}{e^x+1}$ and use $y' = \frac{u'}{1+u^2}$. The expressions u' and $1+u^2$ share the same denominator, which cancels cleanly.

Detailed derivation

1. Let $u = \frac{e^x-1}{e^x+1}$, so $y = \arctan u$ and $y' = \frac{u'}{1+u^2}$.
2. By the quotient rule, $u' = \frac{e^x(e^x+1)-(e^x-1)e^x}{(e^x+1)^2} = \frac{2e^x}{(e^x+1)^2}$.
3. Compute $1+u^2 = \frac{(e^x+1)^2+(e^x-1)^2}{(e^x+1)^2} = \frac{2(e^{2x}+1)}{(e^x+1)^2}$.
4. Dividing gives $y' = \frac{\frac{2e^x}{(e^x+1)^2}}{\frac{2(e^{2x}+1)}{(e^x+1)^2}} = \frac{e^x}{e^{2x}+1}$.
5. Because $e^x+1>0$ and $e^{2x}+1>0$, both the function and its derivative are defined for every $x \in \mathbb{R}$.

Common pitfalls

- using $\frac{1}{1+u}$ as the derivative of $\arctan u$
- losing exponential terms in the quotient rule
- canceling incorrectly before placing $1+u^2$ over a common denominator

Verification

At $x = 0, u = 0$ and $u' = \frac{1}{2}$, so $y'(0) = \frac{1}{2}$. The simplified formula also gives $\frac{1}{1+1} = \frac{1}{2}$.

Method takeaway

After nested differentiation, look for shared structure before expanding; here the denominators of u' and $1 + u^2$ cancel completely.

Extension

Replace e^x by any positive differentiable function $w(x)$ and apply the same structure to arctan$[\frac{w-1}{w+1}]$.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.5, openstax-calculus-v1-3.7, openstax-calculus-v1-3.9. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Classic-method variant

Q086 · Prove the Inverse-Function Derivative Formula from First Principles

Problem

Suppose f is one-to-one on a neighborhood U of x_0, its image contains a neighborhood of $y_0 = f(x_0)$, and it has inverse $x = g(y)$ on that image. Assume f is differentiable at x_0, $f'(x_0) \neq 0$, and $g(y) \to x_0$ as $y \to y_0$. Prove $g'(y_0) = \frac{1}{f'(x_0)}$.

Answer

$g'(y_0) = \frac{1}{f'(x_0)}$

Knowledge points

- corresponding increments for inverse functions
- reciprocal difference quotients
- reciprocal law for a nonzero limit

Reading and method choice

The y-increment for the inverse and the x-increment for the original function describe the same pair of points. Rewrite the quotient for g as the reciprocal of the quotient for f and use $f'(x_0) \neq 0$.

Detailed derivation

1. Let $\Delta y = y - y_0$ and $\Delta x = g(y) - g(y_0) = g(y) - x_0$. Since g is the inverse, $\Delta y = f(x_0 + \Delta x) - f(x_0)$.
2. As $y \to y_0$, the assumption gives $\Delta x \to 0$. For $y \neq y_0$, the single-valued inverse relation also gives $\Delta x \neq 0$.
3. The inverse difference quotient is $\frac{g(y)-g(y_0)}{y-y_0} = \frac{\Delta x}{f(x_0+\Delta x)-f(x_0)}$.
4. Rewrite it as a reciprocal: $\frac{\Delta x}{f(x_0+\Delta x)-f(x_0)} = \frac{f(x_0+\Delta x)-f(x_0)}{\Delta x}^{-1}$.
5. The expression in braces tends to $f'(x_0) \neq 0$ as $\Delta x \to 0$, so the reciprocal tends to $\frac{1}{f'(x_0)}$. Hence $g'(y_0) = \frac{1}{f'(x_0)}$.

Common pitfalls

- confusing an inverse function with a reciprocal function
- not explaining why $\Delta y \to 0$ makes the corresponding $\Delta x \to 0$
- taking a reciprocal even when $f'(x_0) = 0$

Verification

For $f(x) = 3x + 2$, the inverse is $g(y) = \frac{y-2}{3}$. The formula gives $g' = \frac{1}{3}$, agreeing with direct differentiation.

Method takeaway

The inverse derivative formula switches the same local rate from $\frac{\Delta y}{\Delta x}$ to its reciprocal $\frac{\Delta x}{\Delta y}$.

Extension

In an application, first locate $x_0 = g(y_0)$; do not substitute y_0 directly into f'.

Source lineage: Classic-method variant · inverse_exponential_logarithmic_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.7. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Calculation Synthesis Classic-method variant

Q087 · The nth Derivative of a Cubic Times an Exponential

Problem

Let $y = x^3e^{2x}$. For $n \geq 3$, find $y^{(n)}$ with no summation sign in the answer.

Answer

$y^{(n)} = e^{2x}[2^nx^3 + 3n2^{n-1}x^2 + 3n(n-1)2^{n-2}x + n(n-1)(n-2)2^{n-3}]$.

Knowledge points

- Higher derivative of a product
- Termination of polynomial derivatives
- Binomial coefficients

Reading and method choice

Distribute the n differentiations between x^3 and e^{2x}. The fourth derivative of x^3 is zero, so only the cases of assigning 0, 1, 2, or 3 derivatives to the polynomial remain.

Detailed derivation

1. The higher product formula gives $y^{(n)} = \sum_{k=0}^{n} C(n,k)(x^3)^{(k)}(e^{2x})^{(n-k)}$.
2. We have $(x^3)^{(0)} = x^3, (x^3)' = 3x^2, (x^3)'' = 6x, (x^3)''' = 6$, and all terms with $k \geq 4$ vanish.
3. Also, $(e^{2x})^{(m)} = 2^me^{2x}$, so retain only $k = 0, 1, 2, 3$.
4. The four terms are $2^nx^3e^{2x}, 3n2^{n-1}x^2e^{2x}, 6C(n,2)2^{n-2}$xe^{2x}, and $6C(n,3)2^{n-3}e^{2x}$.
5. Use $6C(n,2) = 3n(n-1)$ and $6C(n,3) = n(n-1)(n-2)$, then factor out e^{2x} to obtain the answer.
6. The condition $n \geq 3$ makes all four displayed terms and coefficients directly applicable.

Common pitfalls

- Using 2^k instead of 2^{n-k}
- Omitting the binomial coefficients
- Writing the third derivative of x^3 as 3

Verification

At $n = 3$, the formula gives $e^{2x}(8x^3 + 36x^2 + 36x + 6)$; differentiating $y'' = e^{2x}(4x^3 + 12x^2 + 6x)$ once more gives the same result.

Method takeaway

The nth derivative of a polynomial times an exponential is a finite sum whose number of terms is controlled by the polynomial degree.

Extension

For any cubic polynomial in place of x^3, apply the same structure termwise by linearity.

Source lineage: Classic-method variant · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Proof | Synthesis | Classic-method variant

Q088 · Inductive Proof of the Higher Product Formula

Problem

Suppose u and v have the required derivatives. Prove by induction that for $n \geq 0, (uv)^{(n)} = \sum_{k=0}^{n} C(n,k)u^{(k)}v^{(n-k)}$.

Answer

The formula holds. Differentiate the sum termwise and combine coefficients using $C(n,k) + C(n,k-1) = C(n+1,k)$.

Knowledge points

- Higher product formula
- Mathematical induction
- Binomial coefficient recurrence

Reading and method choice

Differentiating the nth-order formula splits every term into two. After aligning all terms of the form $u^{(k)}v^{(n+1-k)}$, neighboring binomial coefficients combine into the next row.

Detailed derivation

1. For $n = 0$, the right side is the single term $C(0,0)uv = uv$, so the base case holds.
2. Assume $(uv)^{(n)} = \sum_{k=0}^{n} C(n,k)u^{(k)}v^{(n-k)}$.
3. Differentiate termwise and apply the product rule to obtain $\sum_{k=0}^{n} C(n,k)[u^{(k+1)}v^{(n-k)} + u^{(k)}v^{(n+1-k)}]$.
4. Reindex the first sum with $j = k + 1$, which covers $j = 1, \ldots, n+1$; the second covers $j = 0, \ldots, n$.
5. The endpoint coefficients at $j = 0$ and $j = n + 1$ are 1, while each interior coefficient is $C(n, j-1) + C(n, j) = C(n+1, j)$.
6. Hence $(uv)^{(n+1)} = \sum_{j=0}^{n+1} C(n+1, j)u^{(j)}v^{(n+1-j)}$, the desired formula at $n + 1$.
7. Mathematical induction proves the result for all $n \geq 0$.

Common pitfalls

- Failing to change summation bounds after reindexing
- Omitting the two endpoint terms
- Reversing the subscripts in the binomial recurrence

Verification

At $n = 2$, the formula gives $u''v + 2u'v' + uv''$, exactly what results from differentiating the ordinary product rule once more.

Method takeaway

The binomial coefficients in the Leibniz formula count the ways to distribute differentiations between two factors.

Extension

If u is a degree-m polynomial, $u^{(k)} = 0$ for $k > m$, so the formula automatically truncates to at most $m + 1$ terms.

Source lineage: Classic-method variant · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Error diagnosis | Synthesis | Original synthesis / diagnosis

Q089 · Diagnosing a Missing Repeated Chain Factor

Problem

For $y = \ln(2x - 1), x > \frac{1}{2}, a$ student writes $y^{(n)} = \frac{(-1)^{n-1}(n-1)!}{(2x-1)^n}$. Identify the error, give the correct formula, and check at least $n = 1$ and $n = 2$.

Answer

The student misses a factor 2 at every order. The correct formula is $y^{(n)} = \frac{(-1)^{n-1}(n-1)!2^n}{(2x-1)^n}$.

Knowledge points

- Repeated chain factors
- Higher logarithmic derivatives
- Error localization

Reading and method choice

The first derivative contributes one factor 2, and each later differentiation of a negative power of $2x - 1$ contributes another. Thus the nth derivative contains 2^n.

Detailed derivation

1. Directly, $y' = \frac{2}{2x-1}$; the student's $n = 1$ formula gives only $\frac{1}{2x-1}$, revealing the missing factor.
2. Differentiating again gives $y'' = -\frac{4}{(2x-1)^2}$, with another factor 2 from the chain rule.
3. Set $u = 2x - 1$. The nth derivative of $\ln u$ with respect to u is $\frac{(-1)^{n-1}(n-1)!}{u^n}$.
4. Because u is linear, each differentiation with respect to x multiplies the corresponding u-derivative by $\frac{du}{dx} = 2$, giving 2^n in total.
5. Therefore $y^{(n)} = \frac{(-1)^{n-1}(n-1)!2^n}{(2x-1)^n}$.
6. For $n = 1$ and $n = 2$ this gives $\frac{2}{2x-1}$ and $-\frac{4}{(2x-1)^2}$, confirming the formula.

Common pitfalls

- Including the inner derivative only at the first order
- Replacing the accumulated factor 2^n by 2n
- Checking signs and powers but ignoring coefficients

Verification

At $x = 1$, the correct formula gives $y'(1) = 2$; the defining difference quotient for $\ln(1 + 2h)$ also gives 2, while the student's formula gives 1.

Method takeaway

For a linear composite, the same inner slope factor accumulates once per derivative order.

Extension

For $\ln(ax + b)$, the formula is $\frac{(-1)^{n-1}(n-1)!a^n}{(ax+b)^n}$.

Source lineage: Original synthesis / diagnosis · higher_derivatives · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Calculation | Synthesis | Classic-method variant

Q090 · Second Derivative of a Cubic Implicit Curve

Problem

The curve $x^3 + y^3 = 6xy$ passes through $(3,3)$. Find y' and y'' at that point.

Answer

$y' = -1$ **and** $y'' = -\frac{16}{3}$.

Knowledge points

- Differentiating a cubic implicit curve
- Second implicit derivative
- Product and chain rules

Reading and method choice

First obtain the derivative relation and y' at the point. In the second differentiation, both y^2y' and xy' are products.

Detailed derivation

1. Differentiate once: $3x^2 + 3y^2y' = 6y + 6xy'$.
2. At $(3,3)$, $27 + 27y' = 18 + 18y'$, so $y' = -1$.
3. Differentiate again: $6x + 6y(y')^2 + 3y^2y'' = 6y' + 6(y' + xy'')$.
4. The right side simplifies to $12y' + 6xy''$.
5. Substitute $x = 3, y = 3, y' = -1$ to get $18 + 18 + 27y'' = -12 + 18y''$.
6. Thus $9y'' = -48$ and $y'' = -\frac{16}{3}$.

Common pitfalls

- Omitting $6y(y')^2$ from $\frac{d(3y^2y')}{dx}$
- Dropping a $6y'$ term from $\frac{d(6xy')}{dx}$
- Carrying an incorrect sign for y'

Verification

$(3, 3)$ satisfies $27 + 27 = 54$. In the second derivative equation, both sides become -108 after substitution.

Method takeaway

For a complicated second implicit derivative, list every product-rule contribution explicitly.

Extension

The same procedure works at other nonsingular points; if the coefficient of y' vanishes, first inspect the local tangent direction.

Source lineage: Classic-method variant · implicit_differentiation · reference IDs: mit-18.01sc-differentiation, openstax-calculus-v1-3.8. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Classic-method variant

Q091 · Deriving the Second Parametric Derivative Formula

Problem

Let $x = x(t)$ and $y = y(t)$ be twice differentiable with $x'(t) \neq 0$. Prove $\frac{d^2y}{dx^2} = \frac{x'(t)y''(t) - y'(t)x''(t)}{[x'(t)]^3}$.

Answer

Start from $\frac{dy}{dx} = \frac{y'(t)}{x'(t)}$, then use $\frac{d}{dx} = \frac{1}{x'(t)}\frac{d}{dt}$ and the quotient rule.

Knowledge points

- First parametric derivative
- Relation between $\frac{d}{dx}$ and $\frac{d}{dt}$
- Quotient rule
- Second derivative

Reading and method choice

Treat $\frac{dy}{dx}$ as a function of t. Differentiating with respect to x means differentiating in t and then dividing by $\frac{dx}{dt}$.

Detailed derivation

1. Since $x'(t) \neq 0$, $\frac{dy}{dx} = \frac{y'(t)}{x'(t)}$.
2. For any differentiable $H(t)$, $\frac{dH}{dx} = \frac{\frac{dH}{dt}}{\frac{dx}{dt}} = \frac{H'(t)}{x'(t)}$.
3. With $H(t) = \frac{y'(t)}{x'(t)}$, $\frac{d^2y}{dx^2} = \frac{1}{x'}\frac{d}{dt}\left(\frac{y'}{x'}\right)$.
4. The quotient rule gives $\frac{d}{dt}\left(\frac{y'}{x'}\right) = \frac{y''x' - y'x''}{(x')^2}$.
5. Multiplying by $\frac{1}{x'}$ yields $\frac{d^2y}{dx^2} = \frac{x'y'' - y'x''}{(x')^3}$.
6. Every division uses $x'(t) \neq 0$, completing the proof.

Common pitfalls

- Using $\frac{y''(t)}{x''(t)}$
- Reversing the quotient-rule numerator

- Forgetting the final factor $\frac{1}{x'}$
- Omitting $x' \neq 0$

Verification

For $x = t^2 + 1, y = t^3$, the formula gives $\frac{(2t)(6t)-(3t^2)(2)}{(2t)^3} = \frac{3}{4t}$, matching the direct result.

Method takeaway

A second parametric derivative converts variables twice, which creates the cube $[x'(t)]^3$ in the denominator.

Extension

The formula applies only where $x' \neq 0$; points with $x' = 0$ require separate analysis.

Source lineage: Classic-method variant · parametric_differentiation · reference IDs: openstax-calculus-v2-7.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Error diagnosis | Synthesis | Classic-method variant

Q092 · Diagnosing a Frozen Water-Surface Radius

Problem

An inverted conical tank is 9 m high with rim radius 3 m. The water depth rises at 0.2 m min^{-1}. Find the volume rate when $h = 3$ m. A student fixes the water radius at 3 m, writes $V = \frac{1}{3}\pi \cdot 3^2 h$, and obtains $\frac{dV}{dt} = 0.6\pi$ m^3 min^{-1}. Diagnose the error and give the correct result.

Answer

The water-surface radius changes with h and cannot be fixed at 3 m. Since $\frac{r}{h} = \frac{1}{3}$ and $V = \frac{\pi h^3}{27}$, at $h = 3$ m we have $\frac{dV}{dt} = 0.2\pi$ m^3 min^{-1}.

Knowledge points

- Instantaneous water-surface radius
- Similar triangles
- Related-rate error diagnosis
- Cone volume

Reading and method choice

The 3 m radius belongs to the full rim, not to every water depth. The error freezes the time-dependent radius r as a constant.

Detailed derivation

1. The tank height-to-radius ratio is $9 : 3 = 3 : 1$. Similarity gives $\frac{r}{h} = \frac{1}{3}$.
2. Thus the current water radius is $r = \frac{h}{3}$, not always 3 m.
3. Substitute into $V = (\frac{1}{3})\pi r^2 h$ to get $V = (\frac{1}{3})\pi(\frac{h^2}{9})h = \frac{\pi h^3}{27}$.
4. Differentiate: $\frac{dV}{dt} = \frac{(\frac{\pi h^2}{9})dh}{dt}$.
5. At $h = 3$ m and $\frac{dh}{dt} = 0.2$ m min^{-1}, $\frac{dV}{dt} = \left(\frac{\pi \cdot 9}{9}\right)0.2 = 0.2\pi$ m^3 min^{-1}.
6. The student's 0.6π is too large because radius 3 m occurs only when the water reaches $h = 9m$.

Common pitfalls

- Confusing fixed tank dimensions with instantaneous water dimensions
- Freezing a variable by early substitution
- Omitting similarity
- Failing to check the dimension $\mathrm{m}^3\ \mathrm{min}^{-1}$

Verification

At $h = 3$ m, $r = 1$ m. Direct differentiation of $V = \frac{\pi}{3}r^2h$ with $r' = \frac{h'}{3}$ also gives 0.2π.

Method takeaway

A common related-rate error is freezing a geometric quantity that still varies with time; identify every time-dependent variable first.

Extension

For a cylinder, the cross-sectional radius is fixed, so $V = \pi R^2 h$ may be used directly.

Source lineage: Classic-method variant · related_rates · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.1. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Classic-method variant

Q093 · Equivalence between a Linear Increment Decomposition and a Derivative

Problem

Prove that a one-variable function f is differentiable at x_0 if and only if there is a constant A such that $\Delta y = A\Delta x + o(\Delta x)$. Prove that $A = f'(x_0)$, so $dy = Adx$.

Answer

If the derivative exists, set $r = \Delta y - f'(x_0)\Delta x$ and observe $\frac{r}{\Delta x} \to 0$. Conversely, divide the decomposition by Δx to obtain derivative A.

Knowledge points

- Definition of derivative
- Differentiable increment representation
- Little-o notation
- Biconditional proof

Reading and method choice

Prove both directions. Construct a remainder from the derivative in one direction; divide the linear decomposition by Δx in the other.

Detailed derivation

1. Assume first that $f'(x_0)$ exists, so $\lim_{\Delta x\to 0} \frac{\Delta y}{\Delta x} = f'(x_0)$.
2. Define $r(\Delta x) = \Delta y - f'(x_0)\Delta x$. For $\Delta x \neq 0$, $\frac{r}{\Delta x} = \frac{\Delta y}{\Delta x} - f'(x_0) \to 0$.
3. Thus $r = o(\Delta x)$, and $\Delta y = f'(x_0)\Delta x + o(\Delta x)$, with $A = f'(x_0)$.
4. Conversely, suppose $\Delta y = A\Delta x + o(\Delta x)$. For $\Delta x \neq 0$, $\frac{\Delta y}{\Delta x} = A + \frac{o(\Delta x)}{\Delta x}$.
5. As $\Delta x \to 0$, the second term tends to 0, so the derivative exists and $f'(x_0) = A$.
6. Therefore A is unique, and $dy = Adx = f'(x_0)dx$.

Common pitfalls

- Proving only one implication
- Using $o(\Delta x)$ without its ratio definition

- Dividing at $\Delta x = 0$
- Failing to identify A with the derivative

Verification

The two arguments are reverse algebraic processes. Since a limit is unique, the coefficient A is unique.

Method takeaway

A derivative can be described by a difference quotient or by a linear principal part in the increment; the differential is that principal part.

Extension

The same decomposition immediately proves that differentiability implies continuity because $\Delta y \to 0$ as $\Delta x \to 0$.

Source lineage: Classic-method variant · differentials_and_linear_approximation · reference IDs: mit-18.01sc-applications, openstax-calculus-v1-4.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Error diagnosis | Synthesis | Classic-method variant

Q094 · Diagnosing a Wrong Base Point and an Exact Equality

Problem

To approximate $\sqrt{25.5}$, a student takes $f(x) = \sqrt{x}, x_0 = 25$, and $\Delta x = 0.5$, but writes $dy = f'(25.5)\Delta x = \frac{0.5}{2\sqrt{25.5}} \approx 0.04951$ and claims $\sqrt{25.5} = 5.04951$. Identify all key errors, give the correct differential approximation, and state whether a rigorous error bound follows.

Answer

The derivative must be evaluated at the base point $25 : dy = f'(25) \cdot 0.5 = (\frac{1}{10}) \cdot 0.5 = 0.05$, so $\sqrt{25.5} \approx 5.05$. An approximation sign is required, and this method gives no automatic rigorous error bound.

Knowledge points

- Base point of a linearization
- Differential approximation
- Equality versus approximation
- Limits of error claims

Reading and method choice

A linearization is the tangent prediction at the base point, so its slope is $f'(x_0)$, not the slope at the unknown target. The prediction is not an exact equality.

Detailed derivation

1. The formula is $f(x_0 + \Delta x) \approx f(x_0) + f'(x_0)\Delta x$, so the derivative is evaluated at x_0.
2. Here $f(25) = 5$ and $f'(x) = \frac{1}{2\sqrt{x}}$, so $f'(25) = \frac{1}{10}$.
3. The differential is $dy = f'(25)\Delta x = (\frac{1}{10})(0.5) = 0.05$.
4. Therefore $\sqrt{25.5} \approx 5 + 0.05 = 5.05$.
5. Using $f'(25.5)$ is not the stated base-point linearization, and the student also replaces approximation by equality.
6. Differential linearization in this chapter alone does not provide a rigorous error bound; a numerical check is not a remainder proof.

Common pitfalls

- Evaluating the slope at the target instead of the base point
- Treating dy as exact Δy
- Calling a numerical check a rigorous proof
- Failing to distinguish base point and increment

Verification

$5.05^2 = 25.5025$, only 0.0025 from 25.5. The actual $\sqrt{25.5} \approx 5.04975$ is close to 5.05.

Method takeaway

A linearization uses the function value and derivative at the base point; its output is a first-order approximation.

Extension

A closer convenient base point may improve the approximation, but still does not create a rigorous error bound without additional theory.

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Hard | Parameter / synthesis / application | Challenge | Classic-method variant

Q095 · Evaluating the 2026th Derivative at Zero

Problem

Let $y = e^x(\cos x + \sin x)$. Without expanding derivative by derivative, find $y^{(2026)}(0)$ and explain the cyclic structure used.

Answer

$y^{(2026)}(0) = 2^{1013}$.

Knowledge points

- Higher exponential-trigonometric derivatives
- Auxiliary angle
- Phase cycle modulo 8

Reading and method choice

First combine $\cos x + \sin x$ into one phase-shifted cosine. Each differentiation then multiplies the amplitude by $\sqrt{2}$ and advances the phase by $\frac{\pi}{4}$.

Detailed derivation

1. Use $\cos x + \sin x = \sqrt{2}\cos(x - \frac{\pi}{4})$, so $y = \sqrt{2}e^x \cos(x - \frac{\pi}{4})$.
2. For $e^x \cos(x + \varphi)$, every differentiation multiplies the amplitude by $\sqrt{2}$ and adds $\frac{\pi}{4}$ to the phase.
3. Hence $y^{(n)} = \sqrt{2}(\sqrt{2})^n e^x \cos(x - \frac{\pi}{4} + \frac{n\pi}{4})$.
4. Set $n = 2026$ and $x = 0$ to get $y^{(2026)}(0) = 2^{\frac{2027}{2}} \cos(\frac{2025\pi}{4})$.
5. Since $2025 \equiv 1 \pmod{8}$, $\cos\left(\frac{2025\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$.
6. Therefore $y^{(2026)}(0) = 2^{\frac{2027}{2}} \cdot 2^{-\frac{1}{2}} = 2^{1013}$.

Common pitfalls

- Using the wrong sign for the phase shift
- Noticing a phase cycle but ignoring amplitude growth
- Combining the exponent $\frac{2027}{2}$ with the cosine value incorrectly

Verification

At $n = 1$, the formula gives $2e^x \cos x$, exactly matching $y' = e^x(\cos x + \sin x - \sin x + \cos x)$.

Method takeaway

Very high derivatives of exponential-trigonometric products reduce to a fixed amplitude multiplier and a finite phase cycle.

Extension

Use the same formula to compute $y^{(2027)}(0)$ and compare two consecutive values.

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Hard Calculation Challenge Classic-method variant

Q096 · The nth Derivative of a Rational Function with a Repeated Pole

Problem

Let $y = \frac{1}{x^2(x-1)}, x \neq 0, 1$. Find $y^{(n)}$ for $n \geq 0$.

Answer

$y^{(n)} = (-1)^n n![\frac{1}{(x-1)^{n+1}} - \frac{1}{x^{n+1}} - \frac{n+1}{x^{n+2}}]$.

Knowledge points

- Partial fractions with repeated factors
- Higher derivatives of negative integer powers
- Linear combinations

Reading and method choice

Because the denominator contains x^2, the partial fractions must include both $\frac{1}{x}$ and $\frac{1}{x^2}$. After decomposition, use the nth-derivative formulas for reciprocal and reciprocal-square terms.

Detailed derivation

1. Set $\frac{1}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1}$.
2. Multiplying by $x^2(x-1)$ gives $1 = Ax(x-1) + B(x-1) + Cx^2$; coefficient comparison yields $A = -1, B = -1, C = 1$.
3. Thus $y = -\frac{1}{x} - \frac{1}{x^2} + \frac{1}{x-1}$.
4. Use $[(x-a)^{-1}]^{(n)} = (-1)^n n!(x-a)^{-n-1}$.
5. Also, $(x^{-2})^{(n)} = (-1)^n (n+1)! x^{-n-2}$.
6. Combine the three derivatives and factor $(-1)^n n!$ to obtain the stated answer.

Common pitfalls

- Using only one term $\frac{A}{x}$ for a repeated factor
- Dropping the minus sign before $\frac{1}{x^2}$
- Forgetting the factor $n+1$ after factoring $(n+1)!$

Verification

At $n = 0$, the formula is $\frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2}$; combining the terms gives numerator $x^2 - x(x-1) - (x-1) = 1$, recovering the original function.

Method takeaway

A repeated linear factor requires a complete ladder of reciprocal powers in the partial-fraction decomposition.

Extension

A factor $(x-a)^m$ requires every term from $\frac{1}{x-a}$ through $\frac{1}{(x-a)^m}$.

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Challenge Proof Challenge Original synthesis / diagnosis

Q097 · Critical Exponents in a Two-Parameter Oscillatory Family

Problem

Let $\alpha, \beta > 0$ and $f_{\alpha,\beta}(x) = \begin{cases} |x|^\alpha \sin\left(\frac{1}{|x|^\beta}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$ (1) Prove that $f_{\alpha,\beta}$ is continuous at 0. (2) Completely classify differentiability at 0. (3) When it is differentiable, completely classify continuity of $f'_{\alpha,\beta}$ at 0.

Answer

(1) Continuous for all $\alpha, \beta > 0$; (2) differentiable at 0 exactly when $\alpha > 1$, with $f'(0) = 0$; (3) f' is continuous at 0 exactly when $\alpha > \beta + 1$.

Knowledge points

- critical parameter exponents
- control of bounded oscillation
- first-principles differentiation
- chain rule
- special sequences for necessity

Reading and method choice

Continuity is controlled by $|x|^\alpha$. Differentiability is controlled by $\alpha-1$ in the quotient. Differentiating the phase introduces the amplification $|x|^{-\beta-1}$, raising the critical exponent for continuity of the derivative to $\beta + 1$.

Detailed derivation

1. Since $|f_{\alpha,\beta}(x)| \le |x|^\alpha$ and $\alpha > 0$, the right-hand side tends to $0 = f_{\alpha,\beta}(0)$ as $x \to 0$. Thus the function is continuous at 0 for all $\alpha, \beta > 0$.
2. The origin quotient is $\operatorname{sgn}(h)|h|^{\alpha-1}\sin(\frac{1}{|h|^\beta})$. If $\alpha > 1$, its absolute value is at most $|h|^{\alpha-1} \to 0$, so $f'(0) = 0$.
3. If $\alpha = 1$, take positive $h_n = (\frac{1}{\frac{\pi}{2}+2\pi n})^{\frac{1}{\beta}}$, for which the quotient is 1, and $k_n = (\frac{1}{\frac{3\pi}{2}+2\pi n})^{\frac{1}{\beta}}$, for which it is -1. Thus no derivative exists.
4. If $0 < \alpha < 1$, along the same h_n the quotient is $h_n^{\alpha-1} \to +\infty$, so there is no finite derivative. Hence differentiability at 0 holds exactly when $\alpha > 1$.
5. For $x \neq 0$, differentiation gives $f'(x) = \operatorname{sgn}(x)[\alpha|x|^{\alpha-1}\sin(\frac{1}{|x|^\beta}) - \beta|x|^{\alpha-\beta-1}\cos(\frac{1}{|x|^\beta})]$.

6. If $\alpha > \beta + 1$, then $\alpha - 1 > 0$ and $\alpha - \beta - 1 > 0$. The two terms are bounded in absolute value by $\alpha|x|^{\alpha-1}$ and $\beta|x|^{\alpha-\beta-1}$, so both tend to 0 and $f'(x) \to 0 = f'(0)$.

7. If $1 < \alpha \leq \beta + 1$, take $x_n = (\frac{1}{2\pi n})^{\frac{1}{\beta}} > 0$. Then the sine term is 0 and the cosine term is 1, so $f'(x_n) = -\beta x_n^{\alpha-\beta-1}$. It equals $-\beta$ when the exponent is 0 and is unbounded when the exponent is negative.

8. Therefore, within the differentiable range, f' is continuous at 0 exactly when $\alpha > \beta + 1$.

Common pitfalls

- looking only at $|x|^\alpha$ and forgetting that a difference quotient lowers the power by one
- omitting sgn(x) or $\beta + 1$ when differentiating the phase
- claiming necessity from vague oscillation without sequences
- confusing the conditions $\alpha > 1$ and $\alpha > \beta + 1$

Verification

For $(\alpha, \beta) = (2, 1)$, the function is differentiable at 0 but $\alpha = \beta + 1$, so its derivative is not continuous. For $(3, 1)$, $\alpha > \beta + 1$ and the derivative is continuous, confirming that the thresholds differ.

Method takeaway

After each difference quotient or phase differentiation, recompute the net power controlling the oscillation; the strongest amplification determines the critical exponent.

Extension

Replacing sine by cosine and redefining $f(0)$ appropriately produces a related classification problem for continuity, differentiability, and continuity of the derivative.

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Challenge | Proof | Challenge | Original synthesis / diagnosis

Q098 · The nth Derivative of a Squared Logarithm and Harmonic Numbers

Problem

Define $H_0 = 0$ and $H_m = 1 + \frac{1}{2} + \cdots + \frac{1}{m}$ for $m \geq 1$. For $x > 0$, prove by induction that for $n \geq 1$, $[(\ln x)^2]^{(n)} = 2(-1)^{n-1}(n-1)!x^{-n}[\ln x - H_{n-1}]$.

Answer

The formula holds for every $n \geq 1$; the induction step uses $H_n = H_{n-1} + \frac{1}{n}$.

Knowledge points

- Induction for higher derivatives
- Harmonic-number recurrence
- Product and power differentiation

Reading and method choice

The formula contains a factorial, a negative power, and an n-dependent constant. After differentiating in the induction step, the new $\frac{1}{n}$ combines with H_{n-1} to form H_n.

Detailed derivation

1. For $n = 1$, the right side is $\frac{2(\ln x - H_0)}{x} = \frac{2\ln x}{x}$, exactly $[(\ln x)^2]'$, so the base case holds.
2. Assume the formula for n and abbreviate $C = 2(-1)^{n-1}(n-1)!$.
3. Differentiate $Cx^{-n}(\ln x - H_{n-1})$ to obtain $Cx^{-n-1}[1 - n(\ln x - H_{n-1})]$.
4. Rewrite the bracket as $-n[\ln x - H_{n-1} - \frac{1}{n}] = -n[\ln x - H_n]$, using $H_n = H_{n-1} + \frac{1}{n}$.
5. Since $C(-n) = 2(-1)^n n!$, the result is $2(-1)^n n! x^{-n-1}[\ln x - H_n]$.
6. This is precisely the claimed formula with n replaced by $n + 1$, so the induction step holds.
7. Therefore the formula is valid for every integer $n \geq 1$.

Common pitfalls

- Forgetting $H_0 = 0$ in the base case
- Omitting the derivative $\frac{1}{x}$ of $\ln x$ in the product rule
- Replacing $H_{n-1} + \frac{1}{n}$ by H_{n+1}

Verification

At $n = 2$, the formula gives $-2x^{-2}(\ln x - 1) = \frac{2(1-\ln x)}{x^2}$; direct differentiation of $\frac{2\ln x}{x}$ gives the same expression.

Method takeaway

A carefully chosen recurrence can absorb the new constant produced at every differentiation step in a complicated nth-order formula.

Extension

Compute $n = 3$ directly and check how $H_2 = \frac{3}{2}$ appears in the result.

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Challenge Proof Challenge Classic-method variant

Q099 · A General Second-Derivative Formula for a Separable Implicit Equation

Problem

Let φ and ψ be twice differentiable, with $y = y(x)$ satisfying $\varphi(x) + \psi(y) = C$ and $\psi'(y) \neq 0$. Prove $y'' = -\frac{\varphi''(x)}{\psi'(y)} - \frac{[\varphi'(x)]^2\psi''(y)}{[\psi'(y)]^3}$ using only the differentiation rules of this chapter.

Answer

First obtain $y' = -\frac{\varphi'}{\psi'}$, then differentiate $\varphi' + \psi' y' = 0$ again and substitute y'.

Knowledge points

- Second implicit differentiation
- Product and chain rules
- General formula
- Algebraic elimination

Reading and method choice

Differentiating the first relation $\varphi'(x) + \psi'(y)y' = 0$ again is clearer than applying the quotient rule directly to $y' = -\frac{\varphi'}{\psi'}$.

Detailed derivation

1. The first derivative gives $\varphi'(x) + \psi'(y)y' = 0$, hence $y' = -\frac{\varphi'(x)}{\psi'(y)}$.
2. Differentiate the first relation again; $\varphi'(x)$ becomes $\varphi''(x)$.
3. The product and chain rules give $\frac{d[\psi'(y)y']}{dx} = \psi''(y)(y')^2 + \psi'(y)y''$.
4. Thus $\varphi''(x) + \psi''(y)(y')^2 + \psi'(y)y'' = 0$.
5. Solve for $y'' = -\frac{\varphi''(x)+\psi''(y)(y')^2}{\psi'(y)}$.
6. Substitute $(y')^2 = \frac{[\varphi'(x)]^2}{[\psi'(y)]^2}$ and separate the two terms to obtain the stated formula.
7. The condition $\psi'(y) \neq 0$ makes every division valid.

Common pitfalls

- Omitting y' from $\frac{d[\psi'(y)]}{dx}$
- Failing to use the product rule on $\psi'(y)y'$
- Forgetting the square on y'
- Using denominator power 2 instead of 3

Verification

For $\varphi = x^2$ and $\psi = y^2$, the formula gives $y'' = -\frac{x^2+y^2}{y^3}$, exactly as direct second differentiation of a circle does.

Method takeaway

The general second implicit formula comes from differentiating the first relation again; the new key term is $\psi''(y)(y')^2$.

Extension

For specific problems, repeatedly differentiating the relation is usually safer than memorizing the general formula.

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Challenge Proof Challenge Classic-method variant

Q100 · Proving the Product Rule for Differentials from the Definition

Problem

Let $u = u(x)$ and $v = v(x)$ be differentiable at x_0. Using only increment decompositions, prove that uv is differentiable at x_0 and $d(uv) = v\,du + u\,dv$. Do not cite the mean value theorem or later-chapter tools.

Answer

Use $\Delta u = u'\Delta x + o(\Delta x), \Delta v = v'\Delta x + o(\Delta x)$, and $\Delta(uv) = v\Delta u + u\Delta v + \Delta u\Delta v$. The last term is $o(\Delta x)$, leaving linear part $(vu' + uv')\Delta x$.

Knowledge points

- Increment definition of differentiability
- Product-increment identity
- Small-order arithmetic
- Product rule for differentials

Reading and method choice

Do not invoke the derivative product rule. Start from the difference between the new and old product and prove that its increment has a linear principal part.

Detailed derivation

1. At x_0 write $u = u(x_0), v = v(x_0)$. Differentiability gives $\Delta u = u'(x_0)\Delta x + o(\Delta x)$ and $\Delta v = v'(x_0)\Delta x + o(\Delta x)$.
2. The product increment is $\Delta(uv) = (u + \Delta u)(v + \Delta v) - uv = v\Delta u + u\Delta v + \Delta u\Delta v$.
3. Substitution in the first two terms gives $v\Delta u + u\Delta v = [vu'(x_0) + uv'(x_0)]\Delta x + o(\Delta x)$.
4. For the remaining product, $\frac{\Delta u\Delta v}{\Delta x} = (\frac{\Delta u}{\Delta x})\Delta v$. Here $\frac{\Delta u}{\Delta x} \to u'(x_0)$ and $\Delta v \to 0$, so the ratio tends to 0 and $\Delta u\Delta v = o(\Delta x)$.
5. Thus $\Delta(uv) = [vu'(x_0) + uv'(x_0)]\Delta x + o(\Delta x)$, proving uv differentiable at x_0.
6. Its linear principal part is $d(uv) = [vu' + uv']\,dx = v(u'\,dx) + u(v'\,dx) = v\,du + u\,dv$.
7. The argument uses only the differentiability definition, algebraic expansion, and small-order arithmetic.

Common pitfalls

- Circularly citing the derivative product rule
- Dropping $\Delta u\Delta v$

- Failing to show $\Delta u \Delta v = o(\Delta x)$
- Confusing base values with increments

Verification

For $u = x$ and $v = x^2$, the rule gives $d(x^3) = x^2dx + x \cdot 2xdx = 3x^2dx$, matching the power-function differential.

Method takeaway

Differential rules follow from linear principal parts; the product $\Delta u \Delta v$ is absorbed into the higher-order remainder.

Extension

The same method proves sum and difference rules; a quotient rule additionally requires a nonzero denominator value at the base point.

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